A

ASME Specifications

A list of specifications available from the American Society of Mechanical Engineers is available in the section Some Commonly Used Specifications, Codes, Standards, and Texts at the end of the book before the index. This list provides additional detail on items that may not be covered in great depth here.

Abrasives

Abrasives is a term given to various materials with different physical formats (such as aggregate, grains, shot, particles bonded with an adhesive, and so forth) that are used to wear down surfaces to desired dimensions or surface finishes or for some other purpose. They may be used in their "raw" state, such as with shot or glass in shot or glass-bead peening operations. Or they may be used in conjunction with adhesives and fillers to make belts, wheels, and tool surfaces.

Sometimes peening operations address more than surface finish. Glass-bead peening has been used to add a compressive stress layer to the surface of gas-turbine compressor wheels to bring the net tensile stress level down to tolerable levels. As alloy metallurgy improved, machinery component operation totally under the stress endurance curve was possible, and such applications were phased out. They are worth mentioning, however, as they can contribute to puzzling failure modes if they are wrongly applied during design or repair and overhaul.

Abrasives may also be combined with adhesives to make grinding wheels, belts, or other components for precision grinding in sophisticated manufacturing or repair machinery such as a blade-tip grinder. A *blade-tip grinder* is basically a combination of a grinder and a computer or some form of computer numerical control (CNC) that is used to grind airfoil blade tips to arrive at precise turbine-wheel-assembly dimensions.

Product improvements now have produced superabrasives for high-precision processes that are generally used in conjunction with CNC and computer-aided design/computer-aided manufacturing (CAD/CAM) programming and equipment.

Different types of grinding materials allow larger depths of cut (decreased work time) with less metallurgical workpiece (heat soak) damage. This lowers overall production costs.

In the continued drive for improved environmental impact in production, some companies are using water coolants rather than oil coolants.

The applications of abrasives in process technology are too numerous to mention; however, some basic information on the common facets of abrasives in various states and process formats follows. Sources are as acknowledged.

CNC Applications*

Precision-plated grinding wheels are now replacing standard wheels (that use adhesive bonding with abrasive grit) in some applications. These are suitable for

* Source: Abrasive Technology, Inc., USA. Adapted with permission.

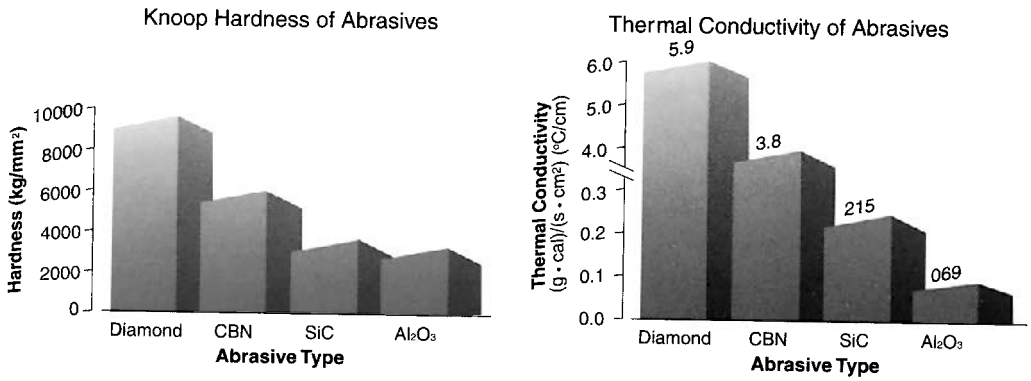


FIG. A-1 Physical characteristics of specific abrasives. (Source: Abrasive Technology, Inc.)



FIG. A-2 CAD/CAM workstation. (Source: Abrasive Technology, Inc.)

use on nickel- and cobalt-based superalloys, engineered ceramics, and standard ferrous alloys. (See Figs. A-1 to A-3.) Advantages of these wheels include:

- The ability to manufacture tight-tolerance complex forms.
- Lower initial cost, compared to vitrified and metal bonded superabrasive wheels.
- Freer cutting, resulting in higher material removal rates, less power, and reduced thermal damage to the workpiece.
- The ability to hold form or profile from first to last cut.
- Reduction or elimination of time associated with dressing, setups, and wheel changes.
- Safe, high-speed operation due to a steel core.
- The ability to strip and replate the core.

Precision techniques are required to measure the finished products, including CNC coordinate measuring machine (CMM) capability with associated CAD/CAM features, laser micrometers, form-scan geometry gauge, optical comparators, and other digital gauges.

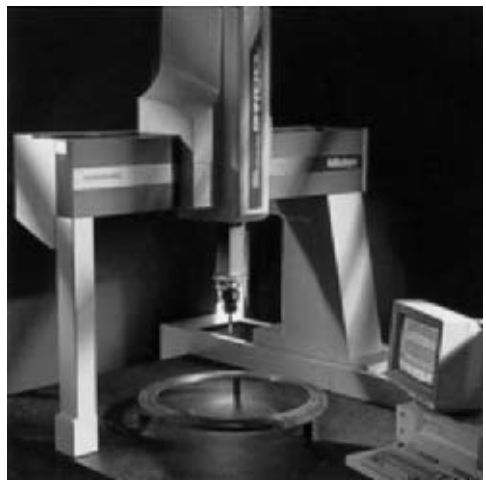


FIG. A-3 Coordinate measuring machine (CMM). (Source: Abrasive Technology, Inc.)

Standard Grinding Wheels*

Technical data standard on abrasives

Wheel symbols and markings. All responsible grinding-wheel manufacturers use a marking system established by the American Standards Association. This marking system describes the makeup of an abrasive wheel in a manner somewhat similar to the way a chemical formula describes the nature of a chemical compound. It indicates by a standard system of letters and numbers the important elements used in the manufacture of the wheel and, to a degree, the amounts and manner of their combination.

Any change in the marking represents a change in wheel characteristics and produces a corresponding change in the action of the wheel.

Familiarity with wheel markings, along with an understanding of the characteristics of the material to be ground, helps to determine the kind of work that a particular wheel is capable of grinding.

Elements of the marking system and their significance. (See Table A-1.) The first "station" specifies the particular type of abrasive material in the wheel. A complete line of abrasive grains suitable for grinding practically any material are available.

The chief difference in the types of abrasive involve their "toughness" and their "friability." The dictionary definition of "friable" is "easily crumbled or reduced to a powder"; certainly abrasives are not easily crumbled or reduced to a powder. The difference is one of degree and both tough and friable types have their applications.

YA	51A	Tough aluminum oxide for heavy-duty work. Best general purpose grain.
AA	52A	Semifriable. Frequently specified for precision grinding on average steels.
TA	53A	Practically pure aluminum oxide. White color, friable, and cool cutting.
	50A	Combination of semifriable and white friable grain.

* Source: Sandusky-Chicago Abrasive Wheel Co., Inc., USA.

TABLE A-1 Summary of Major Abrasive-Type Characteristics

			Sandusky and Chicago	TA 53A	60 60	K K	5 5	V V	E BH			
Sandusky and Chicago Abrasive Types			Grain Sizes	Grades	Structure			Bond Process		Sandusky and Chicago Bond Codes		
	ALUMINUM OXIDE		16	F	2			V—Vitrified			*VITRIFIED*	
					dense						Aluminum Oxide	
YA	Regular	51A	20	G	3						*High heat*	
AA	Semifriable	52A	24	H						R	Red	A
TA	Friable	53A	30	I	4 std.			B—Resin		E	Clear	BH
BB	Off white	–	36	J						E+	Blue	–
32A	Gray	55A	46	K	5					R+	Dark red	–
PA	Pink	12A	60	L	6			R—Rubber			*Low heat*	
					open							
RA	Ruby	RA	70	M	7					–	Red	V
AAT		54A	80	N	8					–	Clear	B
	Combinations											
–		50A	90	O							*Silicon Carbide*	
			100	P	9					A	Clear	C
	SILICON CARBIDE		120	Q	10 very						*RESINOID*	
											Aluminum Oxide	
C	Black	49C	150	R	11 porous					B	Brown	B1
												B2
GC	Green	49CG	180	S	12							BXF
			220	T							*Silicon Carbide*	
										B	Brown	B5
			320	U								B15
											REINFORCED RESIN	
										BF	Brown	BR
											RUBBER	
										HR	Hard	HR
										SR	Soft	SR

AAT	54A	Combination of tough and friable. Used on fairly heavy work but of precision character.
32A	55A	Gray color polycrystalline grain. Excellent for surface and tool and cutter grinders.
PA	12A	Light pink colored grain. Gives superior results on “problem” high alloy steels.
RA	RA	High chrome, ruby colored. Surface grinding and internal wheels.
WTC	WTC	Ceramic abrasive with polycrystalline microstructure. High stock removal rates and long life with vitrified or resin bonds.
ZA	ZA	Zirconium oxide abrasive. High stock removal on mild, stainless, and high alloy steels.
C	49C	Black silicon carbide. Standard for grinding most nonferrous and nonmetallic materials. Best material for use on ordinary cast irons.
GC	49CG	Green silicon carbide. A little more friable than regular black. Usually used in carbide tool grinding.
CA	CA	Combination of aluminum oxide and black silicon carbide and a resin bond. Used in plugs and cones when grinding a variety of materials including both steel and cast iron with the same grinding wheel.

The second "station" specifies the grit size. This represents the approximate number of openings per linear inch in the final screen used to size the grain. Obviously the larger the number, the finer the grain. For example, 60 grit size is one that will pass through a screen with 27 openings per inch and be held on a screen with 33 openings per inch. The most commonly used grit sizes fall in the range of 24 to 120.

The third "station" in the wheel marking is a letter that specifies the grade or relative holding power of the bond that holds the grains in place. With a given type of bond it is the amount used in the wheel that determines the wheel's grade or "hardness." When the amount of bond is increased, the size of the globules of bond (bond posts) connecting each abrasive grain to the adjacent ones is increased. This larger bond post is naturally stronger, thereby increasing the "hardness" of the wheel. The term "hardness" with respect to abrasive wheels has nothing to do with the hardness of the abrasive material itself, but rather with the relative shearing and impact forces necessary to dislodge a grain from the wheel. Both "soft" and "hard" wheels are necessary and have their specific uses.

The ideal grade for any particular grinding application is the one that will hold the grains in place until they become dull or glazed and then release them allowing new sharp grains to take their place. This may be any place on the scale between the very soft G, H, or I grades to the very hard S, T, or U grades depending upon the material being ground, the nature of the results desired, and the various conditions of speed, pressure, etc., coincident to the operation.

The fourth "station" in the marking system describes the "structure" (relative bulk density) of the wheel. When the abrasive grains are tightly pressed together and interlocked the wheel has a denser structure, which is indicated by a lower structure number, such as 3 or 4. When the wheel is rather loosely pressed and has a more porous nature (more free cutting but necessarily less durable), it has a higher structure number, such as 7 or 8.

The standard structure at which most small wheels are manufactured is 4. This standard density is supplied when the structure number is omitted from the wheel marking. Structure numbers higher (more open) than 8 require the addition of a burnout material to the grinding wheel mix. Open structure wheels are soft and should only be used when necessary on difficult to grind materials.

The fifth "station" in the marking system specifies the basic type of bond used in the wheel. These are:

V *Vitrified:* A glass or porcelain fired to a high temperature.
B *A synthetic resin:* Usually a phenolic thermosetting type. ("B" originates from the old term "Bakelite.")
R *Rubber:* Either natural or synthetic oil resistant type.
E *Shellac:* The natural material. Has very limited applications.

The sixth "station" is used to designate the particular bond used in the wheel. These are designed with various characteristics to give the resulting wheel certain physical properties desirable for the different applications.

Fundamentals of wheel selection. There are many factors that enter into the proper selection of a wheel. Probably the most important things to consider when choosing a wheel for the first time on any job are:

A. Material to be ground and its hardness.
B. Amount of stock to be removed.
C. Finish required.
D. Wheel speed or tool speed.

E. Area of grinding contact.
F. Whether grinding is done dry or with a coolant.
G. Severity of the grinding operation.
H. Type and condition of grinding machine being used.

A. The material being ground affects the selection of the type of *abrasive*, the *grit size*, and the *grade*.

Aluminum oxide is used to grind steel and steel alloys. If the material is heat sensitive or a "high-speed" steel, use friable grain. For difficult materials, such as tool steels high in vanadium, pink or ruby grain is suggested.

Silicon carbide is used to grind cast iron, nonferrous metals, and nonmetallic materials such as glass.

Very hard and brittle materials require relatively fine grain, soft or ductile materials a coarser grain.

Use soft grade wheels on hard materials. A harder grade may be used on more easily penetrated materials, which have a lesser dulling effect on the grain.

B. Amount of stock to be removed affects the choice of *grit size* and *bond*.

Use coarse grit for rapid stock removal and finer grits for harder materials. Use vitrified bond for fast cutting and resinoid or rubber for high-speed machines and higher finish.

C. Finish required influences selection of *bond* and *grain size.* Generally speaking a resin-bonded or rubber-bonded wheel gives a better surface finish but not as good dimensional accuracy as a vitrified wheel.

Grain size is a major factor in the surface finish but the method of dressing the wheel also plays an important part. Fine finishes can be obtained with relatively coarse wheels if the wheel is dressed carefully with a diamond and grinding infeed is properly adjusted. A poor finish can result even with a fine wheel if the grinding techniques used are wrong.

A general rule of thumb relating required RMS finish with grit size under average conditions is as follows.

RMS Finish	Grit Size
32	46
20–32	54
15–20	60
10–15	80

D. Wheel speed affects the choice of *bond*. Vitrified wheels are not to be run in excess of 6500 surface feet per minute except in very special cases. High speeds require resin or rubber bonds.

E. Area of grinding contact affects the choice of *grit size* and *grade*. Large contact area indicates coarse grit size and small contact area a finer size. Small contact area requires a harder grade, the smaller the harder.

F. Grinding wet or dry influences the choice of grade. If coolant is used, wheels one or two grades harder may be used without burning the work. Much depends on the amount and efficiency of coolant reaching the area of contact.

G. The severity of the grinding operation influences the choice of *abrasive type* and to some extent the wheel *grade*. A tough abrasive, regular aluminum oxide, would be used for grinding steel under severe conditions. A milder friable abrasive would be used for light grinding on hard or heat-sensitive steels. Intermediate

types, semifriable or a combination, would be used under conditions of average severity. Where rough conditions such as deburring or removing rough scale exist, it is usually necessary to use a harder grade and coarser grit than the nature of the material would call for under normal grinding conditions.

H. The type and condition of the machine being used will influence the choice of *bond* and *grade*. If the machine is of a portable type one may assume that the wheel will have rougher usage than if used on a fixed base machine. If, in addition, it is a high-speed machine, a resin-bonded wheel is mandatory. Generally speaking, portable machines require harder grade wheels than permanently fixed machines.

Peening; Shot Peening*

For illustrative purposes, some standard information on certain types of shot follows.

An example of standard product information

Standard abrasive meets Society of Automotive Engineers (SAE) requirements for size, chemical composition, hardness, microstructure, and physical properties.

Size: See previous chart.

Chemistry

Carbon	.85%–1.20%	
Manganese		
	S70–S110	.35%–1.20%
	S170	.50%–1.20%
	S-230 and larger	.60%–1.20%
Silicon		.40%–1.50%
Sulfur		.05% max
Phosphorus		.05% max

Hardness

S-40/48RC
M 48/55RC
HH 55/62RC
FH 63/68RC

Microstructure: Uniform martensite tempered to a degree consistent with the hardness range with fine, well-distributed carbides, if any.

Physical characteristics: Reference SAE J-827.

Specialty products

Precise media for the most exacting requirements, including roll etch grit for the steel industry, special cutting grit for the granite industry, and military specification shot (MIL-13165-C) for precision peening applications, and customized products to meet a customer's unique requirements.

Packaging

Drums: Nonreturnable 55-gallon steel drums palletized on expendable wooden pallets.

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*Source: National Metal Abrasive, Inc., USA. Adapted with permission.

Shot 2000 lb net
Grit 1667 lb net sizes G-12 through G-40
1500 lb net sizes G-50 through G-120

Bags/cartons: 50 lb net, multiwall paper bags—40 bags per carton (2000 lb net) strapped to disposable wooden pallets.

Bulk: Custom bulk packaging available.

Typical material safety data sheet

SECTION I PRODUCT IDENTIFICATION

D-U-N-S No.: 14-421-8252

Product Names: Perma-Steel Shot, Perma-Steel Grit
Common Name: Cast Steel Shot, Cast Steel Grit
Chemical Family: Ferrous

SECTION II HAZARDOUS INGREDIENTS

CHEMICAL NAME	CAS NUMBER	%WEIGHT	OSHA PEL – TWA	ACGHI TLV – TWA
Iron	1309-37-1	>96	10 mg/m^3 as oxide fume	5 mg/m^3 as oxide fume
Manganese	7439-96-5	0.35–1.3	C* 5 mg/m^3 as dust	C* 5 mg/m^3 as dust
			C* 5 mg/m^3 as fume	C* 1 mg/m^3 as fume
Carbon	1333-86-4	0.8–1.3	None established	None established
Silicon	7440-21-3	0.4–1.2	15 mg/m^3 (as nuisance dust)	10 mg/m^3

SECTION III PHYSICAL DATA

Melting Point: 1371–1482°C
Evaporation Rate: Not Applicable
Boiling Point: 2850–3150°C
Specific Gravity (at 60°F): >7.6
Solubility in Water: Not Applicable
Vapor Pressure: Not Applicable
Vapor Density: Not Applicable
Percent Solid by Weight: 100%
pH: Not Applicable
Appearance and Odor: Shot is near spherical. Grit is angular. Both are gray to blue in color with slight metallic odor.

SECTION IV FIRE AND EXPLOSION HAZARD DATA

Flash Point: Not Applicable
Flammability Limits: Not Applicable
Autoignition Temperature (solid iron exposed to oxygen): 930°C
Cast Steel Shot and Grit will not burn or explode. The solid form of material is not combustible. Fire and explosion hazards are moderate when material is in the form of dust and exposed to heat or flames, chemical reaction, or contact with powerful oxidizer.
Fire Extinguishing Method: Use dry chemicals or sand to exclude air.

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* C means ceiling limit, limits that shall not be exceeded, even for a short time.

Accident Management

Accidents are generally categorized according to levels of severity. These levels would be as follows or subdivisions thereof, depending on the nature of the process. The accident may be one that:

1. Results in death for nonplant personnel (public fatality).
2. Kills one or more plant personnel (major accident).
3. Kills the operator (plant fatality).
4. Injures personnel (time loss) and/or damages property (asset loss).
5. Is prevented by operator's action (near miss).
6. Is prevented by policy/legal area restrictions and personnel procedures and/or manually activated equipment.
7. Is prevented by automatic systems/equipment. The systems may provide passive (monitoring) capability or active real-time control.

When a process engineer provides information to help determine a plant's insurance premiums, he or she will find that those premiums decrease as the accident level goes from 1 to 7 in the previous list. Planning to avoid accidents is thus a major function of the process engineer's duties. Any manual actions/procedures, such as training on plant simulators, required to help avoid accidents should also be practiced or drilled until they become easy to follow when actually required.

To help avoid the possibility of human error, automation in plant-safety systems is now an area of practicing triple redundancy. Triple redundancy was formerly observed only with control of high-performance military aircraft systems. Now, increasingly, it pays off with land-based control systems that are safety, operational, or control in scope. *See* Triple Redundancy.

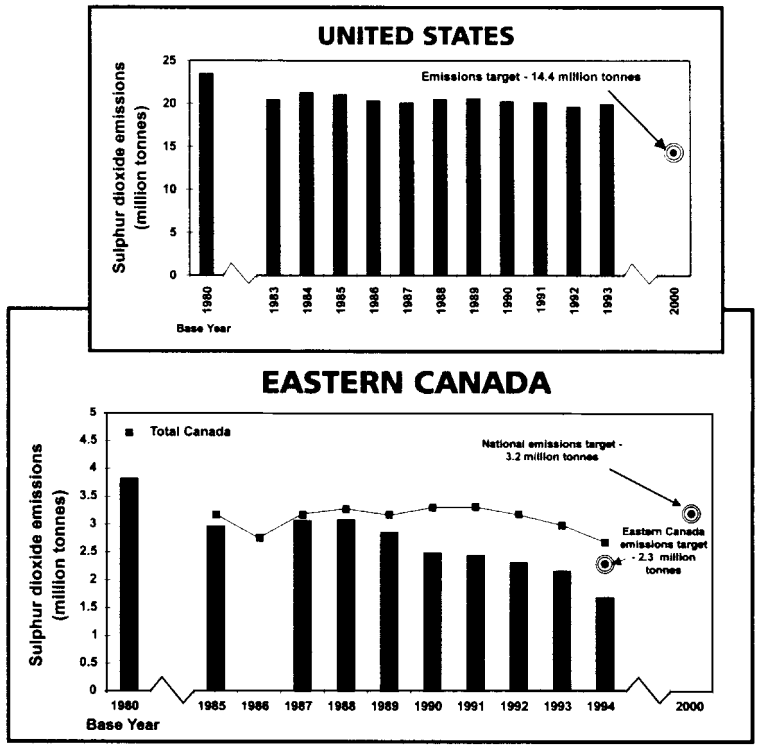
Accountability, Environmental (*see* Environmental Accountability)

Acid Rain

Acid rain is precipitation polluted by acidification with atmospheric pollutants. These pollutants include emissions of oxides of nitrogen (NO_x), oxides of sulfur (SO_x), and hydrogen chloride radicals. Various strengths of nitric acid, sulfuric acid, and hydrochloric acid result. Key indicators of acid rain include emission levels of NO_x and SO_x, wet sulfate deposits, and trends in acidity in lakes and other freshwater bodies. An increase in emissions that increases the level of any of these indicators will bring environmental, regulatory, and potentially public and special-interest group pressures to bear on a plant. Sample measurements of these indicators on a national or provincial scale are illustrated in Figs. A-4 to A-7.

Reference and Additional Reading

1. Soares, C. M., *Environmental Technology and Economics: Sustainable Development in Industry*, Butterworth-Heinemann, 1999.



Notes:
(a) Eastern Canadian emissions data for 1986 are unavailable.
(b) U.S. emission estimates converted to metric tons (1 ton = 0.91 metric tons).

Canadian sources:
Eastern Canadian emissions: Environment Canada. *Annual Report on the Federal-Provincial Agreements for the Eastern Canada Acid Rain Program*, 1990 to 1994.

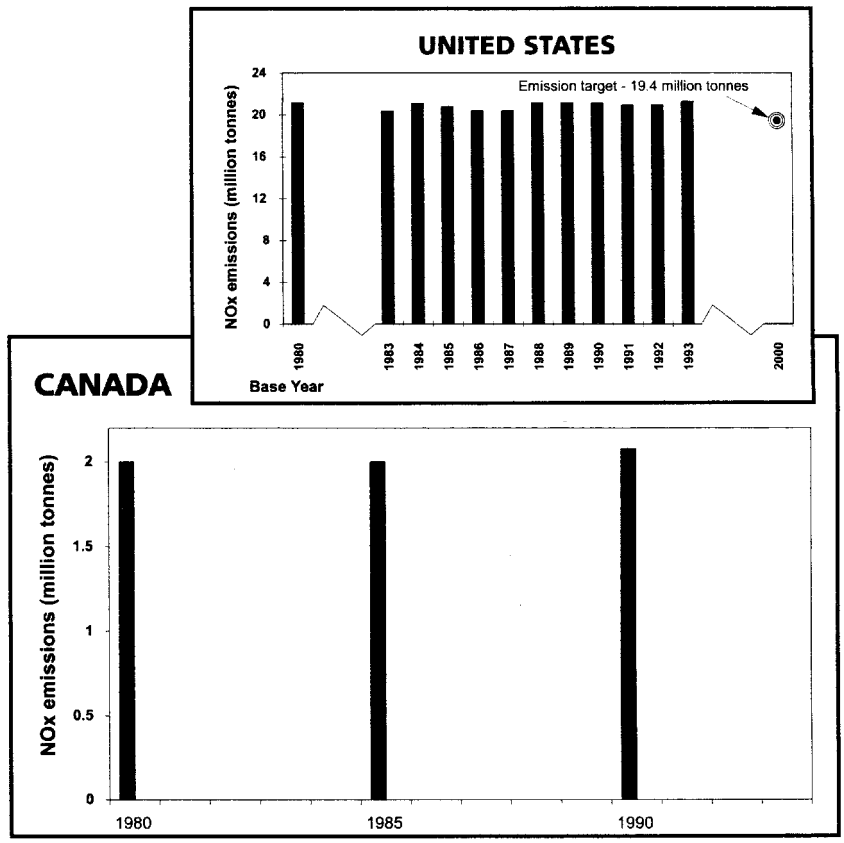
Total Canadian emissions: Pollution Data Branch, Environmental Protection Service, Environment Canada.

U.S. source:
U.S. EPA. 1994. *National Air Pollutant Emission Trends, 1990–1993*. Office of Air Quality Planning and Standards, Research Triangle Park, North Carolina.

FIG. A-4 United States and eastern Canada SO_x emission trends. (Source: Environment Canada SOE 96-2, Spring 1996.)

Acoustic Enclosures, Turbine

Enclosures around noisy rotating machinery, particularly gas turbines, provide protection against noise and help contain risk from situations such as small gas leaks. The design of acoustic gas turbine enclosures is summarized in this section in text extracts from two papers on the subject. These extracts give the reader all the salient points to watch for if specifying or buying an acoustic enclosure. It also provides the reader with basic knowledge of how acoustic sound intensity measurements are conducted generally (i.e., for any kind of equipment). Note also that these designs were built for offshore applications where weight has to be minimized.



Notes:
(a) U.S. emission estimates converted to metric tons (1 ton = 0.91 metric tons).
(b) Canadian emission estimates are preliminary.

Canadian source:
Pollution Data Branch, Environmental Protection Service, Environment Canada.

U.S. source:
U.S. EPA. 1994. *National Air Pollutant Emission Trends, 1990–1993.* Office of Air Quality Planning and Standards, Research Triangle, North Carolina.

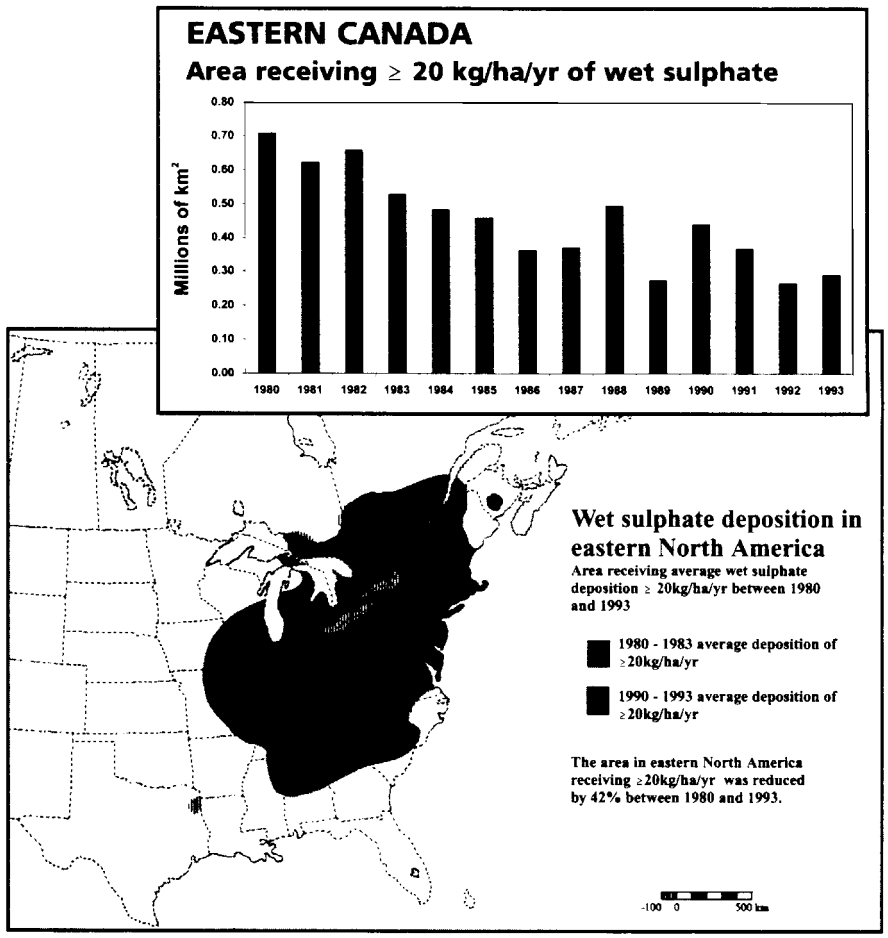
FIG. A-5 United States and Canada acid rain NO_x emission trends. (Source: Environment Canada SOE 96-2, Spring 1996.)

Applications of Sound Intensity Measurements to Gas Turbine Engineering*

Nomenclature

A = total absorption in receiving room
I, I_0 = intensity, pW/m^2
L_1, L_2 = sound pressure levels, dB
LI = sound intensity level, dB

*Source: Altair Filters International Limited, UK; also, this section is adapted from extracts from a paper published in *ASME Journal of Engineering for Gas Turbines and Power*, Vol. 113, October 1991.



Notes:

(a) Wet sulphate deposition represents the weight of sulphate in precipitation and is an indicator of acid rain. The target value of 20 kg/ha/yr of wet sulphate was derived from limited data available in the early 1980s and was based mainly on sport fish loss, which occurs at a pH level of 5.3 and below. More recent studies suggest that a pH of 6.0 is needed to protect most aquatic organisms.

(b) The NAtChem Data Base was used for the storage and analysis of acid precipitation data

(c) National figures on dry sulphate deposition and all nitrate deposition are not available.

Source of data:

Measurements and Analysis Research Division, Atmospheric Environment Service, Environment Canada, Downsview, Ontario.

Map produced by:

State of the Environment Directorate, Environment Canada.

FIG. A-6 Wet sulfate deposits: eastern North America. (Source: Environment Canada SOE 96-2, Spring 1996.)

LK = reactivity index

P, P_0 = pressure, Pa

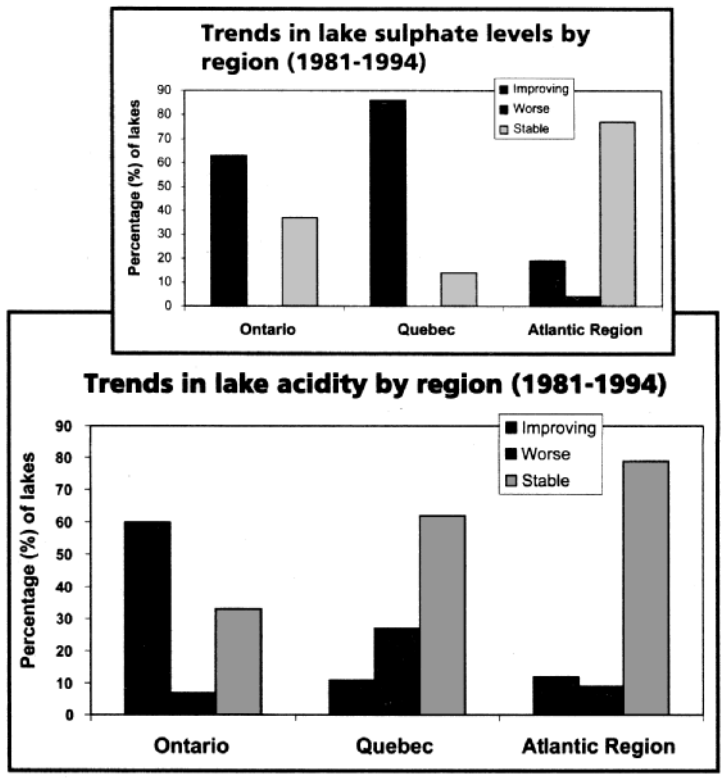
P_a, P_b = pressure, Pa

S = surface area of the test panel, m^2

W, W_0 = power, pW

$r, \Delta r$ = distance, m

t = time, s



Notes:

(a) For acidity: Increasing pH or alkalinity implies an "improving" condition, decreasing pH or alkalinity implies a "worse" condition, no trend means "stable."
(b) For sulphate: Decreasing trends imply "improving," increasing trends imply "worse," no trend means stable.
(c) Number of lakes: Atlantic Region, 75; Quebec, 37; Ontario, 90.
(d) Atlantic Region refers only to the provinces of Nova Scotia and Newfoundland.
(e) Most of the provincial data sets consistently spanned the period from 1981 to 1993/1994. However, some Ontario data sets were slightly longer or shorter than this time period.

Sources:

Methodology: Clair, T.A., *et al.* 1995. "Regional precipitation and surface water chemistry trends in southeastern Canada (1983–1991)." *Canadian Journal of Fisheries and Aquatic Sciences, 52(1): 197–212.*

Data:

Atlantic Region: T.A. Clair, Environment Canada. Quebec: A. Bouchard, Environment Canada. Northwestern Ontario: E.J. Fee, and P. Campbell, Fisheries and Oceans Canada. Algoma: D.S. Jeffries, National Water Research Institute. Sudbury: W. Keller, Ontario Ministry of Environment and Energy. Muskoka/Haliburton: P.J. Dillon, Ontario Ministry of Environment and Energy.

FIG. A-7 Trends in lake sulfate levels (North America). (Source: Environment Canada SOE 96-2, Spring 1996.)

u = particle velocity, m/s

ρ = density, kg/m^3

Gas turbines that are supplied to the oil and power industries are usually given extensive acoustic treatment to reduce the inherent high noise levels to acceptable limits. The cost of this treatment may be a significant proportion of the total cost of the gas turbine installation. In the past it has been difficult to determine if the acoustic treatment is achieving the required noise limits because of a number of

operational problems. These problems include: the presence of other nearby, noisy equipment, the influence of the environment, and instrumentation limitations. Traditionally, sound measurements have been taken using a sound level meter that responds to the total sound pressure at the microphone irrespective of the origin of the sound. So, the enforcement of noise limits has been difficult because of uncertainties concerning the origin of the noise.

Recent advances in signal processing techniques have led to the development of sound intensity meters that can determine the direction, as well as the magnitude, of the sound, without the need for expensive test facilities. These instruments enable the engineer to determine if large equipment, such as gas turbine packages, meet the required noise specification even when tested in the factory or on site where other noise sources are present.

There are, of course, limitations in the use of sound intensity meters, and there are some differences of opinion on measurement techniques. Nevertheless, the acoustic engineer's ability to measure and identify the noise from specific noise sources has been greatly enhanced.

In this section, the differences between sound pressure, sound intensity, and sound power are explained. Measurement techniques are discussed with particular references to the various guidance documents that have been issued. Some case histories of the use of sound intensity meters are presented that include field and laboratory studies relating to gas turbines and other branches of industry.

Fundamental concepts

Sound pressure, sound intensity, and sound power. Any item of equipment that generates noise radiates acoustic energy. The total amount of acoustic energy it radiates is the *sound power*. This is, generally, independent of the environment. What the listener perceives is the *sound pressure* acting on his or her eardrums and it is this parameter that determines the damaging potential of the sound. Unlike the sound power, the sound pressure is very dependent on the environment and the distance from the noise source to the listener.

Traditional acoustic instrumentation, such as sound level meters, detects the sound pressure using a single microphone that responds to the pressure fluctuations incident upon the microphone. Since pressure is a scalar quantity, there is no simple and accurate way that such instrumentation can determine the amount of sound energy radiated by a large source unless the source is tested in a specially built room, such as an echoic or reverberation room, or in the open air away from sound reflecting surfaces. This imposes severe limitations on the usefulness of sound pressure level measurements taken near large equipment that cannot be moved to special acoustic rooms.

Sound intensity is the amount of sound energy radiated per second through a unit area. If a hypothetical surface, or envelope, is fitted around the noise source, then the sound intensity is the number of acoustic watts of energy passing through 1 m^2 of this envelope (see Fig. A-8). The sound intensity, I, normal to the spherical envelope of radius, r, centered on a sound source of acoustic power, W, is given by:

$$I = \frac{W}{4\pi r^2} \tag{1}$$

Clearly, the total sound power is the product of the sound intensity and the total area of the envelope if the sound source radiates uniformly in all directions. Since the intensity is inversely proportional to the distance of the envelope from the noise source, the intensity diminishes as the radius of the envelope increases. But as this

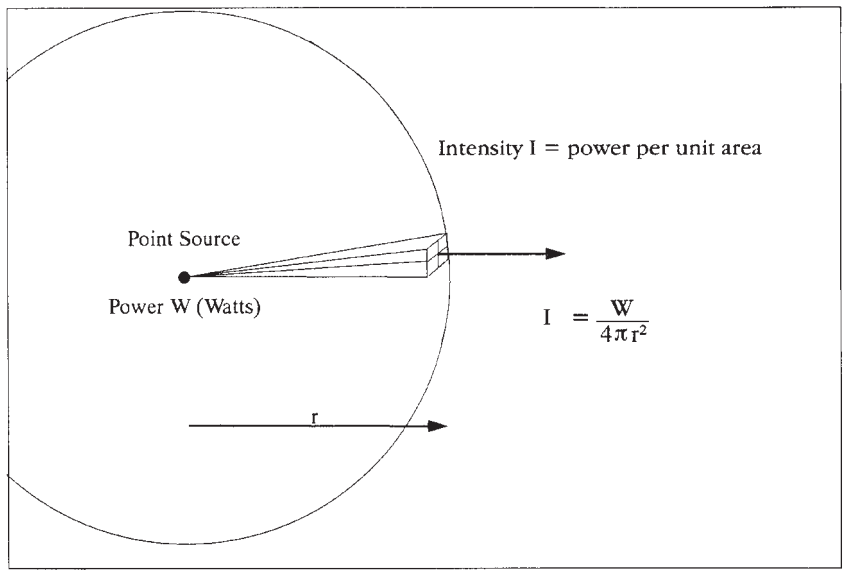


FIG. A-8 The intensity level from a point sound source. (Source: Altair Filters International Limited.)

distance increases, the total area of the envelope increases also, so the product of the intensity and the surface area (equal to the sound power) remains constant.

When a particle of air is displaced from its mean position by a sound wave that is moving through the air there is a temporary increase in pressure. The fact that the air particle has been displaced means that it has velocity. The product of the pressure and the particle velocity is the sound intensity. Since velocity is a vector quantity, so is sound intensity. This means that sound intensity has both direction and magnitude.

It is important to realize that sound intensity is the time-averaged rate of energy flow per unit area. If equal amounts of acoustic energy flow in opposite directions through a hypothetical surface at the same time, then the net intensity at that surface is zero.

Reference levels. Most parameters used in acoustics are expressed in decibels because of the enormous range of absolute levels normally considered. The range of sound pressures that the ear can tolerate is from 2×10^{-5} Pa to 200 Pa. This range is reduced to a manageable size by expressing it in decibels, and is equal to 140 dB.

The sound pressure level (SPL) is defined as:

$$\text{SPL} = 20 \log_{10}\left(\frac{P}{P_0}\right) \quad \text{dB (re. } 2 \times 10^{-5} \text{ Pa)}$$

Likewise, sound intensity level (SIL) and sound power level (PWL) are normally expressed in decibels. In this case,

$$\text{SIL} = 10 \log_{10}\left(\frac{I}{I_0}\right) \quad \text{dB (re. } 1\,\text{pW/m}^2\text{)}$$

$$\text{PWL} = 10 \log_{10}\left(\frac{W}{W_0}\right) \quad \text{dB (re. } 1\,\text{pW)}$$

The relationship between sound pressure level and sound intensity level. When the sound intensity level is measured in a free field in air, then the sound pressure level and sound intensity level in the direction of propagation are numerically the same. In practice most measurements of the sound intensity are not carried out in a free field, in which case there will be a difference between the sound pressure and intensity levels. This difference is an important quantity and is known by several terms, such as reactivity index, pressure–intensity index, *P-I* index, phase index, or LK value. This index is used as a "field indicator" to assess the integrity of a measurement in terms of grades of accuracy or confidence limits. This will be considered in more detail later in this section.

Instrumentation

Sound intensity meters. A sound intensity meter comprises a probe and an analyzer. The analyzer may be of the analog, digital, or FFT (fast Fourier transform) type. The analog type has many practical disadvantages that make it suitable only for surveys and not precision work.

Digital analyzers normally display the results in octave or $^1/_3$ octave frequency bands. They are well suited to detailed investigations of noise sources in the laboratory or on site. Early models tended to be large and heavy and require electrical main supplies, but the latest models are much more suited to site investigations.

FFT analyzers generate spectral lines on a screen. This can make the display very difficult to interpret during survey sweeps because of the amount of detail presented. Another disadvantage of FFT-based systems is that their resolution is generally inadequate for the synthesis of $^1/_3$ octave band spectra.

Sound intensity probes. There are several probe designs that employ either a number of pressure microphones in various configurations or a combination of a pressure microphone and a particle velocity detector. The first type of probe uses nominally identical pressure transducers that are placed close together. Various arrangements have been used with the microphones either side by side, face to face, or back to back. Each configuration has its own advantages and disadvantages.

If the output signals of two microphones are given by P_a and P_b, then the average pressure, P, between the two microphones is:

$$P = {}^1\!/_2\,(P_a + P_b) \tag{2}$$

The particle velocity, u, is derived from the pressure gradient between the two microphones by the relationship:

$$u = -\int \frac{1}{\rho} \cdot \frac{\partial p}{\partial r} \cdot dt$$

$$u = -\frac{1}{\rho} \int \frac{(P_b - P_a)}{\Delta r} \cdot dt \tag{3}$$

Since sound intensity, I, is the product of the pressure and particle velocity, combining equations 2 and 3 gives the intensity as

$$I = -\left(\frac{P_a + P_b}{2\rho\Delta r}\right) \int (P_b - P_a) \cdot dt \tag{4}$$

Figure A-9 shows a two microphone probe, with a face-to-face arrangement, aligned parallel to a sound field. In this orientation the pressure difference is

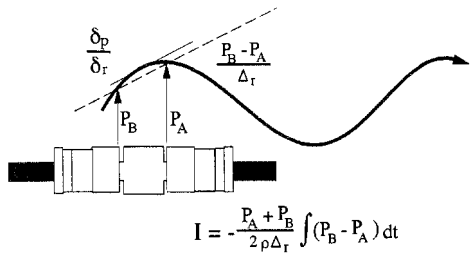


FIG. A-9 The finite difference approximation of sound intensity for a two microphone configuration. (Source: Altair Filters International Limited.)

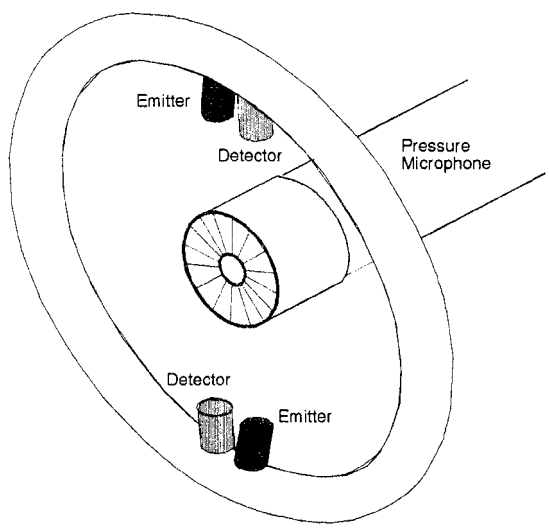


FIG. A-10 Schematic representation of a pressure/velocity probe. (Source: Altair Filters International Limited.)

maximized, and so is the intensity. If the probe is aligned so that the axis of the two microphones is normal to the direction of propagation of the sound wave, then the outputs of the two microphones would be identical in magnitude and phase. Since the particle velocity is related to the difference between the two pressures, P_a and P_b, then the intensity would be zero.

The second type of probe combines a microphone, to measure the pressure, and an ultrasonic particle velocity transducer. Two parallel ultrasonic beams are sent in opposite directions as shown in Fig. A-10. The oscillatory motion of the air caused by audio-frequency sound waves produces a phase difference between the two ultrasonic waves at their respective detectors. This phase difference is related to the particle velocity component in the direction of the beams. This measure of particle velocity is multiplied directly by the pressure to give the sound intensity.

Guidelines and standards in sound intensity measurements and measurement technique

Guidelines and standards. Work began in 1983 on the development of an international standard on the use of sound intensity and the final document is about to be issued. Further standards are expected dealing specifically with instrumentation.

TABLE A-2 Uncertainty of the Determination of Sound Power Level (ISO/DP 9614)

Octave Band Center Frequencies, Hz	1/3 Octave Band Center Frequencies, Hz	Standard Deviations, dB		
		Class 1	Class 2	Class 3
63–125	50–160	2	3	4
250–500	200–630	1.5	2	4
1000–4000	800–5000	1	1.5	4
	6300	2	2.5	4
A-Weighted (50–6300 Hz)		1	1.5	4

NOTES:
1. Class 1 = Precision Grade, Class 2 = Engineering Grade, Class 3 = Survey Grade.
2. The width of the 95% confidence intervals corresponds approximately to four times the dB values in this table.

In the absence of a full standard the only guidance available was the draft ISO standard (ISO/DP 9614) and a proposed Scandinavian standard (DS F88/146).

The ISO document ISO/DP 9614 specifies methods for determining the sound power levels of noise sources within specific ranges of uncertainty.

The proposed test conditions are less restrictive than those required by the International Standards series ISO 3740-3747, which are based on sound pressure measurements. The proposed standard is based on the sampling of the intensity normal to a measurement surface at discrete points on this surface. The method can be applied to most noise sources that emit noise that is stationary in time and it does not require special purpose test environments.

The draft document defines three grades of accuracy with specified levels of uncertainty for each grade. Since the level of uncertainty in the measurements is related to the source noise field, the background noise field, and the sampling and measurement procedures, initial procedures are proposed that determine the accuracy of the measurements. These procedures evaluate the “Field Indicators” that indicate the quality of the sound power measurements. These field indicators consider, among other things:

- The pressure–intensity index (or reactivity index)
- The variation of the normal sound intensities over the range of the measurement points
- The temporal variation of the pressure level at certain monitoring points

The three grades of measurement accuracy specified in ISO/DP 9614, and the associated levels of uncertainty, are given in Table A-2.

The Scandinavian proposed standard (DSF 88/146) was developed for the determination of the sound power of a sound source under its normal operating conditions and in situ. The method uses the scanning technique whereby the intensity probe is moved slowly over a defined surface while the signal analyzer time-averages the measured quantity during the scanning period.

The results of a series of field trials by several Scandinavian organizations suggested that the accuracy of this proposed standard is compatible with the “Engineering Grade,” as defined in the ISO 3740 series.

The equipment under test is divided into a convenient number of subareas that are selected to enable a well-controlled probe sweep over the subarea. Guidance is given on the sweep rate and the line density. Measurement accuracy is graded according to the global pressure–intensity index, LK. This is the numerical difference between the sound intensity level and the sound pressure level. If this

field indicator is less than or equal to 10 dB then the results are considered to meet the engineering grade of measurement accuracy. As this field indicator increases in value, the level of uncertainty in the intensity measurement increases. When the LK value lies between 10 and 15 dB the measurement accuracy meets the "Survey" grade.

Measurement techniques. The precise measurement technique adopted in a particular situation depends on the objectives of the investigation and the level of measurement uncertainty that is required.

(A) Subareas. It was mentioned earlier that the total sound power is the product of the intensity and the surface area of the measurement envelope around the noise source. In practice, most noise sources do not radiate energy uniformly in all directions so it is good practice to divide the sound source envelope into several subareas. Each subarea is then assessed separately, taking into account its area and the corresponding intensity level. The subarea sound powers can then be combined to give the total sound power of the source.

The number, shape, and size of each subarea is normally dictated by two considerations: the physical shape of the source and the variations in intensity over the complete envelope. Subareas are normally selected to conform to components of the whole source such that the intensity over the subarea is reasonably constant. It is important that the subareas are contiguous and the measurement envelope totally encloses the source under investigation.

(B) Sweep or point measurement. Should one measure the intensity levels at discrete positions, with the probe stationary, or should the probe be swept over the subarea? This controversy has occupied much discussion time among practicing acousticians. For precision grade measurements, discrete points are used, but for lower grade work, sweeping is acceptable.

If discrete points are used then the number and distribution of the measurement points must be considered in relation to the field indicators.

In surroundings that are not highly reverberant and where extraneous noise levels are lower than the levels from the source under investigation, relatively few discrete points may be used, distributed uniformly over the surface. The distance from the source may be as great as 1 m.

As the extraneous noise levels increase and/or the environment becomes more reverberant, measurements must be made progressively closer to the source in order to maintain an acceptable level of uncertainty in the measurements. This also requires more measurement points to be used because of the increase in the spatial variation of the intensity distribution.

If sweeping is used then other factors must also be considered. The speed with which the probe is swept across the subarea must be uniform, at about 300 mm/sec, and the area should be covered by a whole number of sweeps with an equal separation between sweep lines. Care must be taken that excessive dwell time does not occur at the edges of the subarea when the probe's direction of sweep is reversed. The operator must also be careful that his or her body does not influence the measurements by obscuring sound entering the measurement area as he or she sweeps.

(C) Distance between source and probe. Generally, the greater the extraneous noise and the more reverberant the environment then the closer should be the probe to the source. In extreme cases the probe may be only a few centimeters from the source surface in order to improve the signal-to-noise ratio. This is normally frowned upon when using conventional sound level meters because measurements

of sound pressure, taken close to a surface, may bear little relation to the pressures occurring further away from the surface. This discrepancy is not due simply to the attenuation with distance that normally occurs in acoustics.

The region very close to a surface is called the "near field." In this region the local variations in sound pressure may be very complex because some of the sound energy may circulate within this near field and not escape to the "far field." This recirculating energy is known as the *reactive* sound field. The sound energy that does propagate away from the surface is called the *active* sound field because this is the component that is responsible for the acoustic energy in the far field.

Since sound intensity meters can differentiate between the active and reactive sound fields, measurements of intensity taken close to noise sources can faithfully indicate the radiated sound energy. However, using a conventional sound level meter near to a noise source may indicate higher sound power levels than occur in the far field because these instruments cannot differentiate between active and reactive fields.

Some advantages and limitations in sound intensity measurements

Background noise. One of the main advantages of the sound intensity method of measurement is that accurate assessments of sound power can be made even in relatively high levels of background noise. But this is only true if the background noise is steady (i.e., not time varying). Using conventional sound pressure level methods the background noise level should be 10 dB below the signal level of interest.

Using sound intensity techniques the sound power of a source can be measured to an accuracy of 1 dB even when the background noise is 10 dB *higher* than the source noise of interest. Figure A-11*a* shows a noisy machine enclosed by a measurement surface. If the background noise is steady, and there is no sound absorption within the measurement surface, then the total sound power emitted by the machine will pass through the measurement surface, as shown.

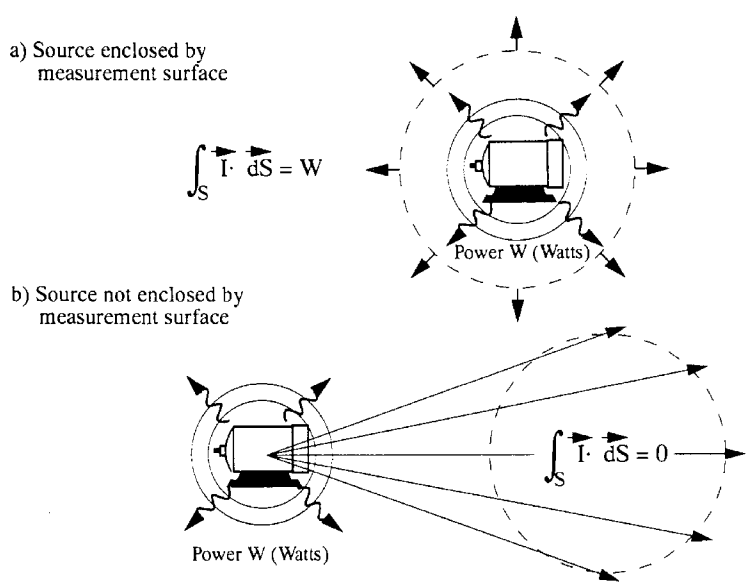


FIG. A-11 The effect of sound sources inside and outside the measurement surface. (Source: Altair Filters International Limited.)

If, however, the noisy machine is outside the measurement surface, as shown in Fig. A-11*b*, then the sound energy flowing into the surface on the left hand side will be emitted from the right hand side of the measurement surface. When the sound intensity is assessed over the whole measurement surface the net sound power radiated from the total surface will be zero.

Effects of the environment. When the sound power of a noise source is evaluated in the field using sound pressure level techniques, it is necessary to apply a correction to the measured levels to account for the effects of the environment. This environmental correction accounts for the influence of undesired sound reflections from room boundaries and nearby objects.

Since a sound intensity survey sums the energy over a closed measurement surface centered on the source of interest, the effects of the environment are cancelled out in the summation process in the same way that background noise is eliminated. This means that, within reasonable limits, sound power measurements can be made in the normal operating environment even when the machine under investigation is surrounded by similar machines that are also operating.

Sound source location. Since a sound intensity probe has strong directional characteristics there is a plane at 90° to the axis of the probe in which the probe is very insensitive. A sound source just forward of this plane will indicate positive intensity, whereas if it is just behind this plane the intensity will be negative (Fig. A-12).

This property of the probe can be used to identify noise sources in many practical situations. The normal procedure is to perform an initial survey of the noise source to determine its total sound power. The probe is pointed toward the source system to identify areas of high sound intensity. Then the probe is reoriented to lie parallel to the measurement surface and the scan is repeated. As the probe moves across a dominant source the intensity vector will flip to the opposite direction.

Testing of panels. The traditional procedure for measuring the transmission loss, or sound reduction index, of building components is described in the series of

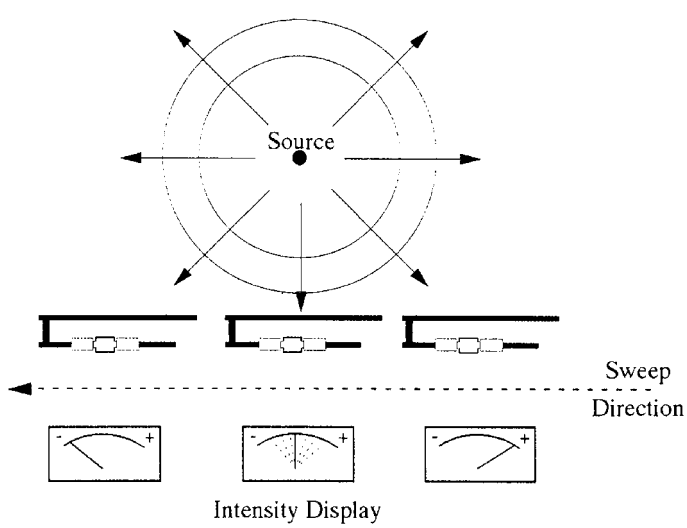


FIG. A-12 Sound source location using the intensity probe. (Source: Altair Filters International Limited.)

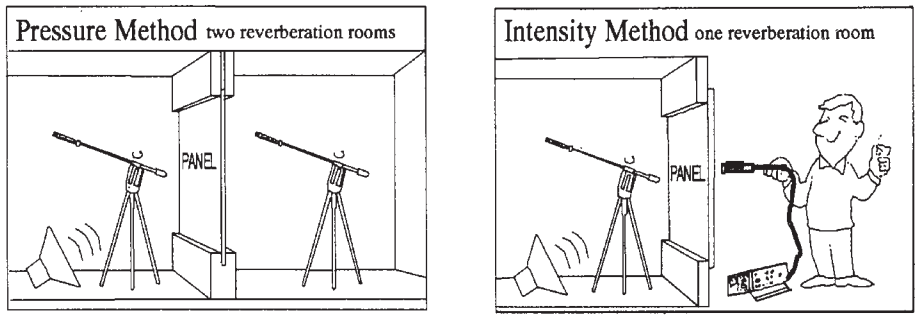


FIG. A-13 Comparison of the pressure and intensity methods for measuring the sound reduction index of panels. (Source: Altair Filters International Limited.)

standards ISO 140. The test method requires the panel under investigation to be placed in an opening between two independent, structurally isolated reverberation rooms, as shown in Fig. A-13.

Sound is generated in the left hand room and the sound pressure levels in the two rooms are measured. Assuming that the sound energy in the right hand room comes through the panel then the sound reduction index (SRI) of the panel is given by:

$$\text{SRI} = L_1 - L_2 + 10 * \log_{10}(S/A)\,\text{dB} \tag{5}$$

For this method to give accurate results, flanking transmission (sound bypassing the test panel) must be minimal and both rooms must be highly reverberant.

If the measurements in the right hand room are carried out using sound intensity, then it is necessary to reduce the amount of reverberation in this room. Since the probe can measure the sound intensity coming through the panel then flanking transmission is no longer a limitation.

For these reasons one can dispense with the second reverberation room altogether. The sound reduction index is then given by:

$$\text{SRI} = L_1 - LI - 6\,\text{dB} \tag{6}$$

If the panel contains a weak area, such as a window, the sound reduction index of the window can be assessed separately. But this will only work if the panel is a greater sound insulator than the window.

Measuring tonal noise sources. Measuring the sound power of tonal noise sources presents difficulties using traditional techniques (ISO 3740, 1980). Unfortunately, using sound intensity techniques on such sources is also fraught with problems. This is because the spatial distribution of the intensity is very sensitive to small alterations in source position and the presence of nearby sound reflective objects.

Case studies of the use of sound intensity

Gas turbine package witness testing. Gas turbine packages are normally assembled in large factory buildings or in the open air between factory buildings. In either case, the environment is totally unsuitable for reliable acoustic tests to be carried out using sound level meters alone. Sound intensity techniques are especially relevant in these situations because of the location in which the tests are to be carried out and because some components, such as the compressor test loop, may

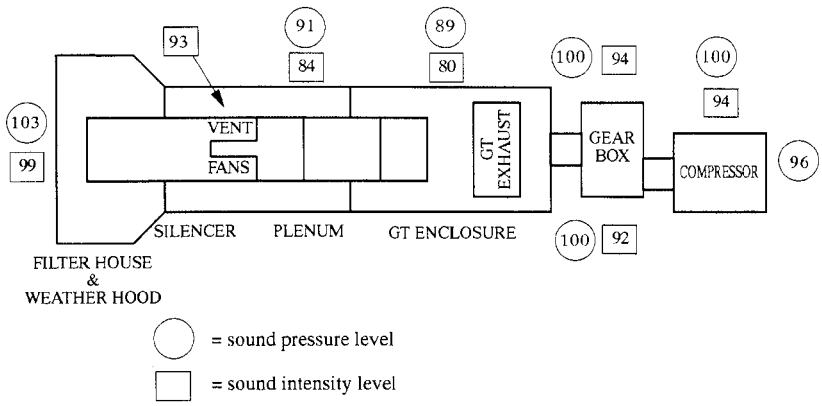


FIG. A-14 Sound pressure and sound intensity levels for a gas turbine package during witness testing. (Source: Altair Filters International Limited.)

TABLE A-3 Rank Ordering of Components in Terms of the A-Weighted Sound Power Levels

Description of Measured Item	Measured Sound Power Levels, dB(A)
Combustion air intake	110
Combustion air plenum and silencer	104
Turbine comp. vent. air breakout	104
Turbine enclosure	103
Compressor casing	103
Gearbox	103
Breakout from temporary exhaust	96

not be contract items. By surveying each component with a sound intensity meter the sound power for each component can be determined separately.

Figure A-14 shows a typical gas turbine driving a compressor. The figures on the drawing indicate the sound pressure and sound intensity levels that were measured during a particular witness test on an RB211 gas turbine package. The sound pressure levels in close proximity to the package were between 89 and 103 dB(A), with the higher levels dominating. Even so, reliable values of the intensity levels were obtained from which the sound power levels were determined. These values are given in Table A-3. Since the sound intensity level is numerically equal to the sound pressure level in free field, the average sound intensity over a given surface area of a gas turbine package provides a direct indication of the average sound pressure level from that surface in free field conditions.

Referring again to Fig. A-14, the sound intensity level measured by the casing of the ventilation fan was 94 dB(A). It would not normally be possible to measure the output from this fan accurately, using a sound level meter, in this situation because of the relatively high sound pressure level in this area due to other sources.

During the testing of another package, the sound power levels from the ventilation fan casing and the fan motor were measured separately. The fan motor was found to be noisier than the manufacturer's stated levels. Discussions with the motor manufacturer revealed that the wrong cooling fans had been fitted to the motors, which accounted for this increase in noise level. The correct cooling fans were subsequently fitted.

TABLE A-4 Comparison of the Sound Levels from Two Roller Mills

Description	Machine A, dB(A)	Machine B, dB(A)
Total sound power for machine	96	94
Sound pressure level by machine	92	89
Sound power level of the motor	93	85

This example clearly illustrates the benefits of sound intensity measurements to check compliance with noise specifications when the test items are very large and are sited in acoustically undesirable areas.

Comparison of two nominally identical production machines. This example is taken from an extensive survey of a production department that had many, relatively small, machines close together in a highly reverberant factory room. The two machines were nominally identical roller mills, as used in many production lines in the paint, flour, and confectionary industries. The drive motors were situated on the top of the machines. A routine sound intensity survey was carried out on each machine during normal production because it was not possible to run the machines in isolation. The two machines are identified as machine A and machine B.

Table A-4 gives the overall, A-weighted sound pressure levels and sound power levels for each machine, and the sound power levels of the motors. The total sound power levels of the machines agreed very well with the values obtained by the manufacturer using sound pressure measurements to derive the sound power levels (ISO 3740, 1980). This technique can only give the total sound power of a machine; it cannot obtain the sound power levels of parts of a machine.

At the time of the survey two machines were each in areas of high noise levels and their total sound power levels were 96 and 94 dB(A). But comparing the motor sound power levels revealed a difference of 8 dB(A) in their respective levels even though both motors were classified as the "low noise" type and cost more than the standard motors. This is just one example where a significant degree of noise control might be achievable by selecting the correct one of two nominally identical electric motors. However, using the traditional method of sound pressure level measurement would not reveal any difference between the motors.

The acoustical properties of flexible connectors. Heavy-duty flexible connectors are used to join separate components of a gas turbine package. When high-performance acoustic hardware is used it is imperative that these flexible connectors do not compromise the total acoustic performance of the package. This is particularly so in the gas turbine exhaust system where multilayered flexible connectors are exposed to high temperatures and severe buffeting from exhaust gases. In some exhaust systems overlapping metal plates are inserted inside the flexible connectors to reduce the buffeting of the flexible material. If additional sound attenuation is required, a heavy, fibrous mat, or "bolster," is inserted between the plates and the flexible connector.

For the acoustics engineer, these flexible connectors are a problem because there is very little data on their acoustical performance, and the designs do not lend themselves to simple theoretical prediction. A brief laboratory investigation was carried out to compare the performances of seven types of flexible connectors with and without plates and bolsters. The tests were carried out in Altair's acoustical laboratory, which was designed to test materials using sound intensity techniques. The samples were physically quite small so the low-frequency performances were

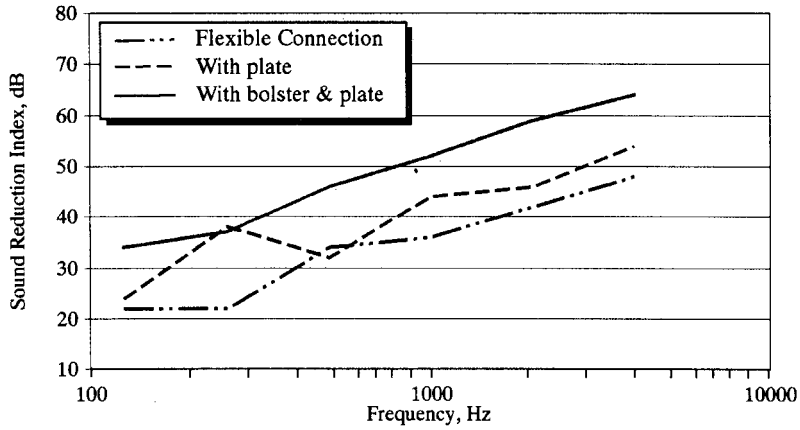


FIG. A-15 Sound reduction index of a flexible connection. (Source: Altair Filters International Limited.)

probably distorted by the small size of the samples. Nevertheless, the exercise yielded much valuable information and led to a simple engineering method of predicting a flexible connector's acoustic performance from a knowledge of its basic parameters.

Figure A-15 compares the sound reduction indices of a typical, multilayered flexible connector tested alone, with plates and with bolster and plates. All of the test results showed a sharp increase in performance at 250 Hz due to the plates. No satisfactory explanation can be offered at this stage for this effect, which may be related to the small size of the samples. But in the middle and high frequencies the results were generally as expected with the plate giving an additional attenuation of about 5 dB compared to the compensator alone. The combination of the plate and bolster gave an additional attenuation of between 10 and 15 dB compared to the flexible connector alone.

Sound source location. During two noise surveys the sound power levels from two different designs of lube oil console were measured using the sound intensity meter. The overall sound power levels were 93 dB(A) and 109 dB(A). The two consoles had electrically driven pumps and both emitted strong tonal noise. In the first case the tonal noise was centered on 8 kHz; using the sound intensity probe it was possible to "home in" on the pump suction pipe as a major noise source. This was confirmed by vibration velocity measurements.

In the second survey the pump outlet pipe gave the highest sound intensity reading. Since this was a relatively long pipe, which was rigidly attached to the frame of the console, it was a dominant source both in its own right and because it was "exciting" the framework. In these cases the sound intensity meter was a useful tool in identifying dominant sources among a large number of small, closely packed noise radiators.

This section has discussed the concepts of sound intensity, sound power, and sound pressure. It has shown how sound intensity meters have given the acoustics engineer a very powerful diagnostic tool. Noise specifications for large, complex machinery can now be checked without the need for special acoustics rooms.

The advantages, and limitations, of sound intensity have been discussed in some detail and several applications have been illustrated by case histories taken from surveys carried out in the process and gas turbine industries.

The superiority of sound intensity meters over sound level meters is clearly apparent. Certain types of laboratory studies can also be carried out more cost effectively using intensity techniques.

Although there are some limitations in the use of sound intensity instrumentation, when used intelligently, it can yield valuable information on dominant noise sources, which, in turn, should provide more cost-effective solutions to noise control in the oil and power industries.

Acoustic Design of Lightweight Gas Turbine Enclosures*

Nomenclature

a, b = panel dimensions, m
B, Bx, By, Bxy = bending and torsional stiffnesses
c = speed of sound in air, m/s
d = fiber diameter, μm
f = frequency, Hz
$f1$ = first panel resonance, Hz
f_c, f_{cx}, f_{cy} = coincidence frequency, Hz
l = thickness of absorptive layer, m
ln = natural logarithm
m = mass per unit area, kg/m^2
$R1$ = flow resistivity index or transmission loss, dB
R = sound reduction
S = stiffness
α = attenuation constant for the material, dB/m
η = damping
λ_m = wavelength of sound in the absorptive layer, m
υ = Poisson ratio
ρ_0 = density of air
ρ_m = density of absorptive layer
τ = transmission coefficient
ω = angular frequency

Gas turbines are used extensively in onshore and offshore environments for power generation, but their use introduces a number of potential hazards. To reduce the risks caused by fire and high noise levels, enclosures, with intake and exhaust silencers, are fitted around the turbines. These enclosures and silencers must be capable of withstanding large static loads produced by equipment sited on top of them and large dynamic loads due to wind.

Traditionally these enclosures are heavy and expensive, especially when stainless steel or aluminum is required for offshore use. This has led to a consideration of more cost-effective designs that still comply with the stringent demands of the oil and gas industry.

One approach that is proving successful is the use of a corrugated enclosure design, which employs a thinner steel wall than its flat panel counterpart, without compromising the structural and fire protection requirements. However, corrugated designs are intrinsically less effective as sound insulators than flat panels. These weaknesses must be understood so that multilayered panels based on the corrugated design can compensate for the deficiencies of unlined, corrugated panels.

This section presents the results of theoretical predictions and measurements on flat and corrugated panels, which were tested in the unlined condition and then with a sound absorbent lining. The effects of varying the profile of the corrugations is also considered.

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* Source: Altair Filters International Limited, UK; also, this section is adapted from extracts from a paper published in *ASME Journal of Engineering for Gas Turbines and Power*, Vol. 113, October 1991.

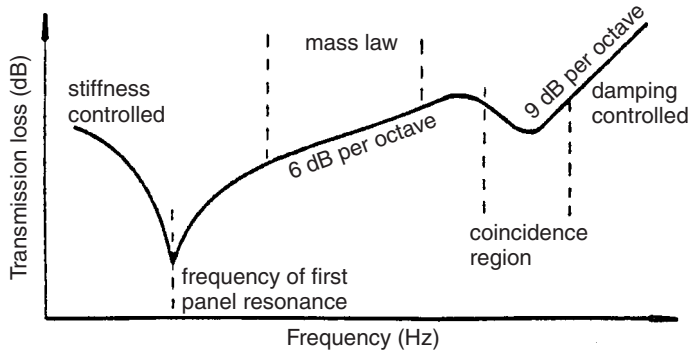


FIG. A-16 Sound reduction index for a typical flat, unlined panel. (Source: Altair Filters International Limited.)

This section considers first the behavior of flat, unlined panels, then describes the physical reasons why corrugated, unlined panels have a different acoustic response to flat panels. The effects of sound-absorbent linings on flat and corrugated panels are then considered.

Sound transmission through unlined flat panels

When a sound wave is incident on a wall or partition, some of the sound energy is transmitted through the wall. The fraction of incident energy that is transmitted is called the transmission coefficient. The accepted index of sound transmission is the sound reduction index, which is sometimes called the transmission loss. This is related to the transmission coefficient by the equation:

$$R = 10 \log_{10}(1/\tau), \text{dB} \tag{1}$$

The behavior of flat panels has been described extensively in the literature, so only the outline of their theoretical performance is given here. The general behavior of a single skin, isotropic panel is shown in Fig. A-16. This characteristic behavior is valid for a wide range of materials, including steel and aluminum.

The propagation of audio-frequency waves through panels and walls is primarily due to the excitation of bending waves, which are a combination of shear and compressional waves. When a panel is of finite extent then a number of resonances are set up in the panel that are dependent on the bending stiffness of the panel. The frequency of the first panel resonance, $f1$, is given by

$$f1 = (\pi/2)\sqrt{(B/m)(1/a^2 + 1/b^2)}, \text{Hz} \tag{2}$$

The second resonance occurs at the "coincidence" frequency, fc, when the projected wavelength of the incident sound coincides with the wavelength of the bending wave in the panel. The coincidence frequency is given by:

$$f_c = (c^2/2\pi)\sqrt{(m/B)}, \text{Hz} \tag{3}$$

The amount of sound transmitted by a panel is dependent on the surface weight, the damping of the panel, and the frequency of the sound. For a finite panel exposed to a random noise field, the acoustic behavior is specified mathematically as follows:

$$R = 20\log_{10} S - 20\log_{10} f - 20\log_{10}(4\pi\rho_0 c), \text{dB}, f < f1 \tag{4}$$

TABLE A-5 Comparison of Predicted and Measured Sound Reduction Indices of Flat, Unlined Panels

Material	Measured or Predicted	Sound Reduction Index, dB							
		63	125	250	500	1000	2000	4000	8000
3-mm steel	Predicted	16	22	28	34	40	43	35	44
	Measured	16	21	27	33	38	39	33	—
6-mm steel	Predicted	22	28	34	40	43	35	44	53
	Measured	22	27	35	39	44	37	42	—
6-mm aluminum	Predicted	13	19	25	31	34	26	35	44
	Measured	13	19	25	30	36	30	32	—

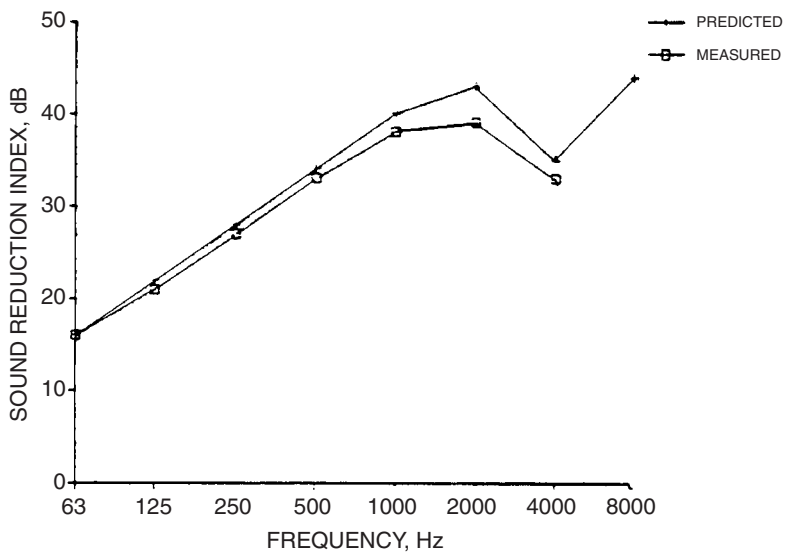


FIG. A-17 Predicted and measured sound reduction indices for 3-mm steel. (Source: Altair Filters International Limited.)

$$R = 20\log_{10}(mf) - 47, \text{dB}, f1 < f < f_c \tag{5}$$

$$R = 20\log_{10}(\pi f m/\rho_0 c) + 10\log_{10}(2\eta f/\pi f_c), \text{dB}, f > f_c \tag{6}$$

In most practical situations the lowest resonance frequency is below the audio range. Above this frequency a broad frequency range occurs in which the transmission loss is controlled by the surface weight and increases with frequency at the rate of 6 dB per octave. In the coincidence region the transmission loss is limited by the damping of the panel. Above the coincidence frequency the transmission loss increases by 9 dB per octave and is determined by the surface weight and the damping.

Clearly for a high value of sound reduction index over the majority of the audio-frequency range (the mass-controlled region) it is better to have a high surface weight, a low bending stiffness, and a high internal damping.

The acoustic performances of some materials have been predicted and are compared to measured values in Table A-5. The data for 3-mm steel and 6-mm aluminum are shown in Figs. A-17 and A-18. Each of the examples shown has the characteristic shape described above and the agreement between the measured and predicted values of transmission loss is good.

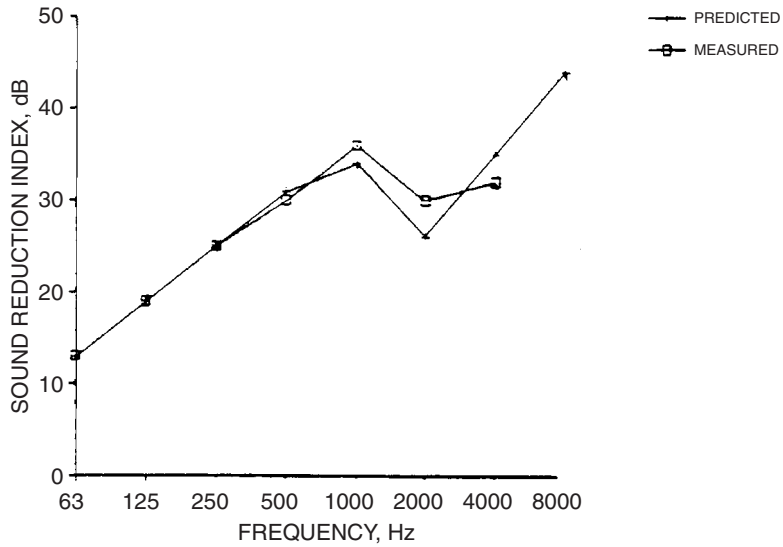


FIG. A-18 Predicted and measured sound reduction indices for 6-mm aluminum. (Source: Altair Filters International Limited.)

Sound transmission through unlined corrugated panels

In corrugated panels the characteristics of the panel are not the same in all directions. The moment of inertia across the corrugations differs from that parallel to the corrugations; thus the bending stiffness varies with direction. This affects both the first panel resonance and the coincidence frequency so that the sound transmission characteristics for these panels differs from flat panels with the same thickness.

The first panel resonance is now given by:

$$f1 = (\pi/2m^{0.5}) \times \sqrt{(Bx/a^4(1-v^2) + By/b^4(1-v^2) + Bxy/a^2b^2)}, \text{Hz} \quad (7)$$

where Bx and By are the bending stiffnesses in the two principal planes of the panel and Bxy accounts for the torsional rigidity of the plate.

For real panels, measuring several meters in length and height, the frequency of this first resonance may still be in the subaudio range but can be several octaves higher than the first resonance of a flat panel of the same dimensions.

Since the coincidence frequency is determined by the bending stiffness, the presence of two bending stiffnesses gives rise to two critical frequencies, f_{cx} and f_{cv}, where:

$$f_{cx} = (c^2/2\pi)\sqrt{(m/Bx)},\ f_{cv} = (c^2/2\pi)\sqrt{(m/By)}$$

If the ratio of the bending stiffnesses, Bx and By, is less than 1.4, then the effects on the transmission loss of the panel will be small, but in typical panels the ratio of the bending stiffnesses is usually much greater than 1.4. This gives rise to a plateau in the transmission loss curve, which is illustrated in Fig. A-19. The plateau may extend over several decades for common corrugated or ribbed panels.

The sound transmission through orthotropic (corrugated) panels has been investigated by Heckl, who derived the following relationships for the diffuse field sound transmission:

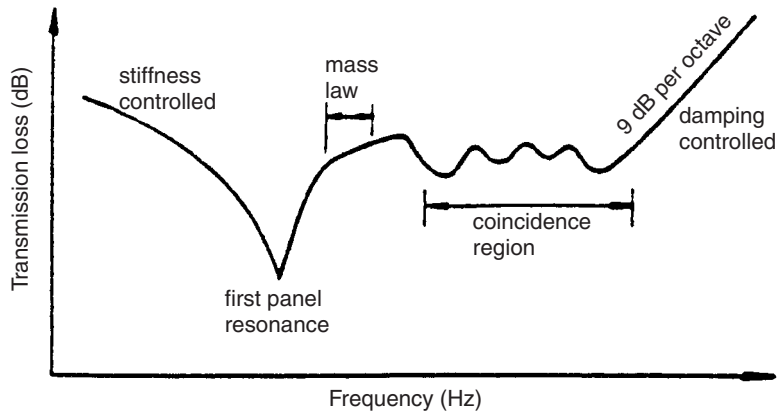


FIG. A-19 Sound reduction index for a typical corrugated, unlined panel. (Source: Altair Filters International Limited.)

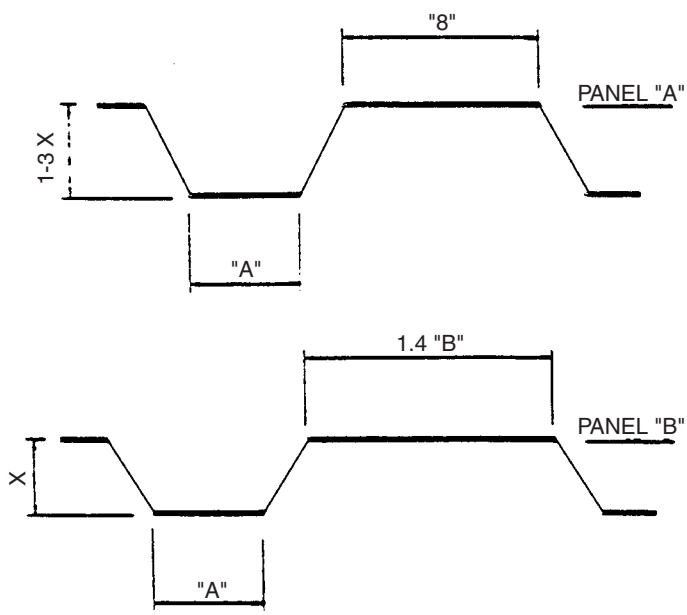


FIG. A-20 Two designs of corrugated panels. (Source: Altair Filters International Limited.)

$$\tau = (\rho_0 c f_{cx}/\pi\omega mf)(\ln(4f/f_{cx}))^2, f_{cx} < f < f_{cy} \quad (8)$$

$$\tau = (\pi\rho_0 c/\omega mf)(f_{cx}f_{cy})^{0.5}, f > f_{cy} \quad (9)$$

where f_{cx} and f_{cy} are the two coincidence frequencies and where f_{cx} is the lower of the two values.

The performances of two designs of corrugated panels have been predicted for panels made of 2.5-mm-thick steel with the designs shown in Fig. A-20. The predicted transmission losses are given in Table A-6 and Fig. A-21.

The predicted panel bending stiffnesses and resonant frequencies for the two panels are tabulated in Table A-7 for a simply supported panel 4 m wide by 2.5 m high. The values for a flat panel of the same overall dimensions have also been

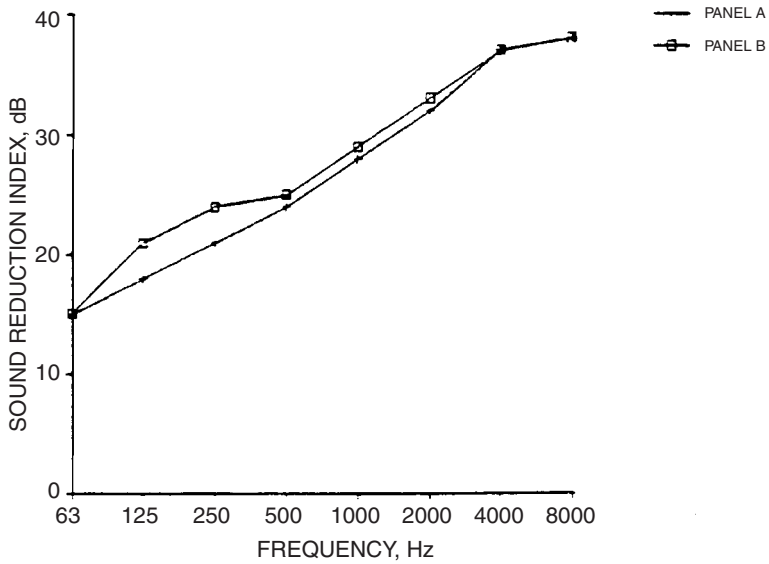


FIG. A-21 Predicted transmission loss for two designs of corrugated panels. (Source: Altair Filters International Limited.)

TABLE A-6 Comparison of Predicted Sound Reduction Indices of Two, Unlined Corrugated Panels

	Sound Reduction Index, dB							
	63	125	250	500	1000	2000	4000	8000
Panel A	15	18	21	24	28	32	37	38
Panel B	15	21	24	25	29	33	37	38

TABLE A-7 Predicted Parameters for Three Panels (the panels were made of steel measuring 4 m by 2.5 m by 2.5 mm thick)

	Bending Stiffness, N·m	First Panel Resonance, Hz	Coincidence Frequency, Hz
Panel A (corrugated)	240,000 : 241	30	163 : 5115
Panel B (corrugated)	90,000 : 263	18	264 : 4897
Flat panel	296	2.9	4825

included. The bending stiffness of panel B, in the direction parallel to the corrugations, is slightly less than that for panel A. This causes a shift in the lower critical frequency of about half an octave, which gives rise to a slightly higher transmission loss at lower frequencies.

Transmission loss measurements have been carried out on a partition with the design of panel A. The test was carried out in accordance with the standard ISO 140 (1978). Table A-8 and Fig. A-22 compare the predicted and measured performances. Both sets of curves show a plateau effect, which is more evident in the measured values. The predicted and measured values are within 4 dB of each

TABLE A-8 Predicted and Measured Sound Reduction Indices for Panel A

	Sound Reduction Index, dB							
	63	125	250	500	1000	2000	4000	8000
Predicted	15	18	21	24	28	32	37	38
Measured	20	23	25	23	23	28	29	29

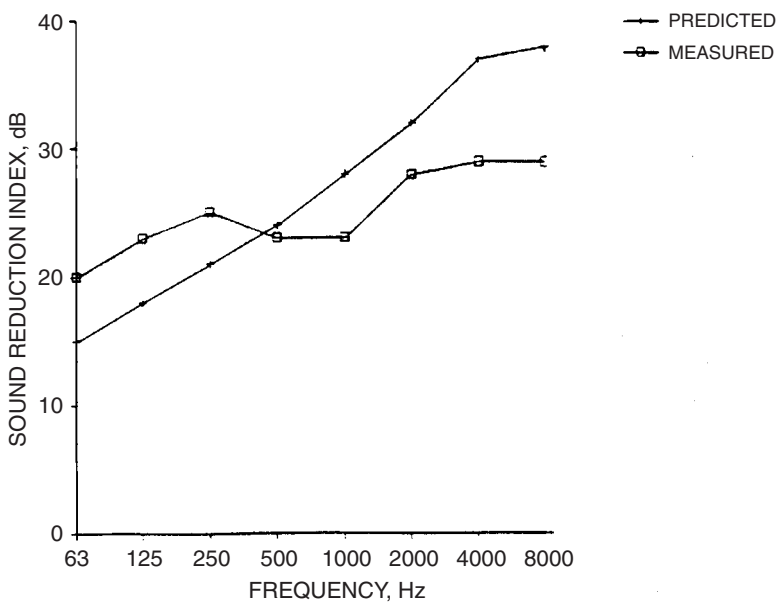


FIG. A-22 Predicted and measured sound reduction indices for panel A. (Source: Altair Filters International Limited.)

other up to 4 kHz, above which the curves differ by about 7 dB. Although the agreement is not as close as for flat panels, it is encouraging. Of more interest is the comparison between the transmission losses of unlined, flat, and corrugated panels with the same thickness. Figure A-23 and Table A-9 show that the corrugated panel is substantially less effective in reducing sound transmission in the mid-frequencies compared to the flat panel equivalent.

Clearly, if noise control is a primary consideration then unlined corrugated panels are not recommended, unless other engineering considerations dictate their use. Corrugated panels offer a considerable increase in bending stiffness, compared to flat panels, which reduces the amount of additional stiffening that would be required for a flat panel. But the volume of material required to produce a corrugated panel is significantly greater than that for a flat panel. For the two corrugated panel designs considered above, panel A contains 23 percent more material than the flat panel, and panel B contains 13 percent more material. Depending on the structural requirements, the additional stiffening required for a flat panel may result in a material savings when a corrugated design is chosen, especially if the flat panel is much thicker to compensate for its inherent low bending stiffness.

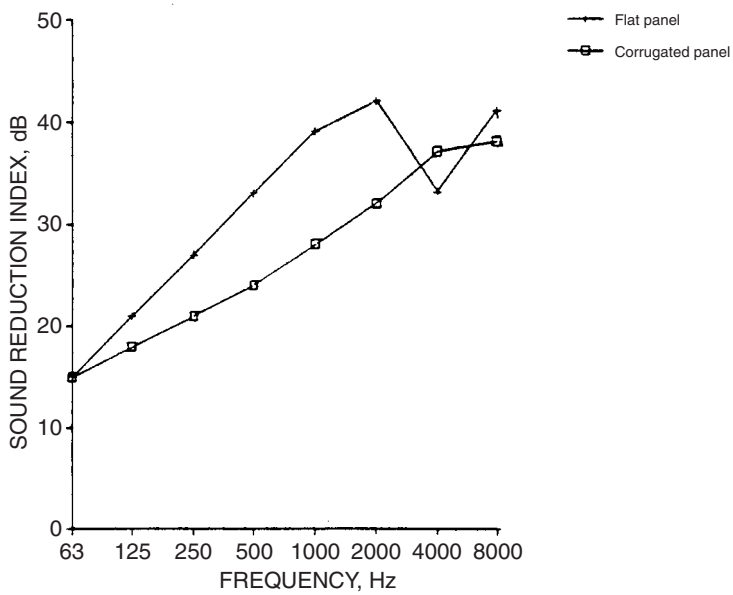


FIG. A-23 Predicted performance of flat and corrugated panels. (Source: Altair Filters International Limited.)

TABLE A-9 Predicted Sound Reduction Indices for Flat Corrugated Panels (Panel Thickness of 2.5 mm)

	Sound Reduction Index, dB							
	63	125	250	500	1000	2000	4000	8000
Flat panel	15	21	27	33	39	42	33	41
Corrugated panel	15	18	21	24	28	32	37	38

The attenuation of sound by absorptive linings

Sound-absorbent linings are frequently fitted in acoustic enclosures to reduce the buildup of reverberant noise inside the enclosure. Typical reductions in the reverberant noise level may be between 3 and 10 dB depending on the application. An additional benefit is the increase in transmission loss of the enclosure panel, which further reduces the noise level outside the enclosure.

The increase in panel transmission loss arises because of several mechanisms. First, if the absorbent lining is sufficiently heavy and the panel is relatively thin, then the added layer may give sufficient additional weight to affect the "mass-law" performance and increase the damping of the panel. At high frequencies the absorbent may be relatively thick in comparison to the wavelength of the sound. The high-frequency sound may be attenuated not only because of the impedance mismatch between the air and the absorbent, but as the sound wave passes through the added layer a significant amount of acoustic energy is converted into heat by viscous losses in the interstices. In practice, the amount of heat generated is minute.

It is possible to distinguish between three frequency regions in which different attenuating mechanisms are predominant. For convenience these are described as

TABLE A-10 Comparison of Predicted and Measured Sound Reduction Indices of Three Lined Panels

	Measured or Predicted	Sound Reduction Index, dB							
		63	125	250	500	1000	2000	4000	8000
Panel 1 (flat)	Predicted	14	22	31	39	48	47	65	63
	Measured	20	21	27	38	48	58	67	66
Panel 2 (flat)	Predicted	20	30	40	46	52	60	63	79
	Measured	31	34	35	44	54	63	62	68
Panel 3 (flat)	Predicted	16	19	22	27	36	44	52	56
(corrugated)	Measured	22	24	28	32	38	48	52	52

Panel 1: 1.6 mm flat steel lined with 100 mm thick glass fiber, 49 kg/cu.m density.
Panel 2: 5 mm flat steel lined with 100 mm thick glass fiber, 48 kg/cu.m density.
Panel 3: 2.5 mm corrugated steel lined with 50 mm thick mineral wool, 64 kg/cu.m density.

regions A, C, and B, where A is the low-frequency region, C is the high-frequency region, and B is the transition region. The boundaries between these three regions are defined by the physical characteristics of the absorptive material in terms of the flow resistivity and the material thickness.

The flow resistivity of fibrous absorptive materials is dependent upon the bulk density and fiber diameter by the approximate relationship:

$$R1 = (3.18 \times 10^3)(\rho_m^{1.53}/d^2)$$

The frequency limits of the three regions, A, B, and C, are defined by:

Region A: $101 \prec \lambda_m$
Region B: $101 \succ \lambda_m$, $\alpha l \prec 9$ dB
Region C: $\alpha l \succ 9$ dB

The values of λ_m, the wavelength of the sound inside the absorptive layer, and α, the attenuation constant for the material, can be measured or predicted for semi-rigid materials:

$$\alpha = (\omega/c)(0.189(\rho_0 f/R1)^{-0.595}) \tag{11}$$

$$\lambda_m = (c/f)(1 + 0.0978(\rho_0 f/R1)^{-0.7})^{-1} \tag{12}$$

For an absorptive layer of known thickness and flow resistivity, the attenuation predicted from the equations given above is additive to that produced by the unlined panel.

The predicted and measured acoustic performances of two flat panels and one corrugated panel, each with an absorptive lining, are shown in Table A-10 and Figs. A-24 to A-26.

The predicted performances of the two flat panels are in good agreement with the measured performances over the majority of the frequency range. The largest discrepancies occur at 63 Hz and 8 kHz.

The agreement between the theoretical and measured performances of the corrugated panel is not as good as for the flat panel. The largest discrepancies occur at the lower frequencies with better agreement occurring at high frequencies. This follows the low-frequency trend shown in Fig. A-23 where the corrugated panel was unlined and the predicted performance was less than the measured performance by 5 dB. Nevertheless, the agreement is sufficiently close to support the theoretical model.

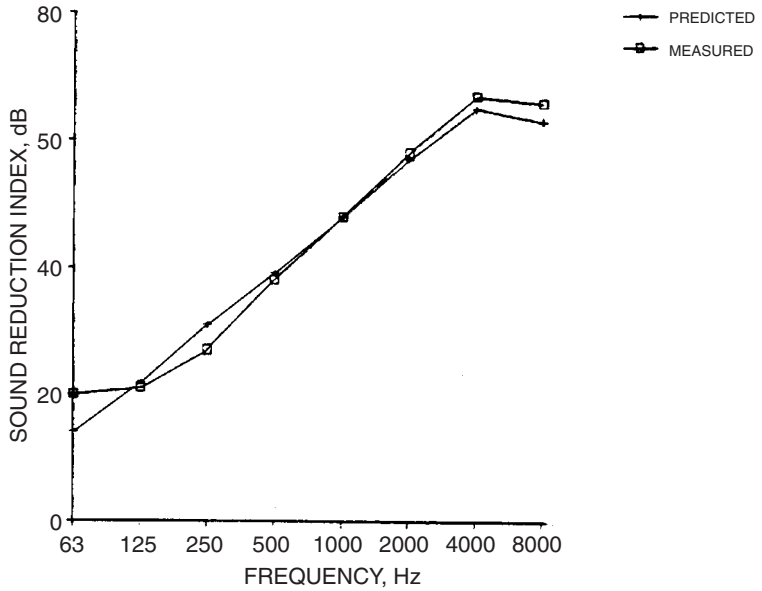


FIG. A-24 Predicted and measured sound reduction index of panel 1. (Source: Altair Filters International Limited.)

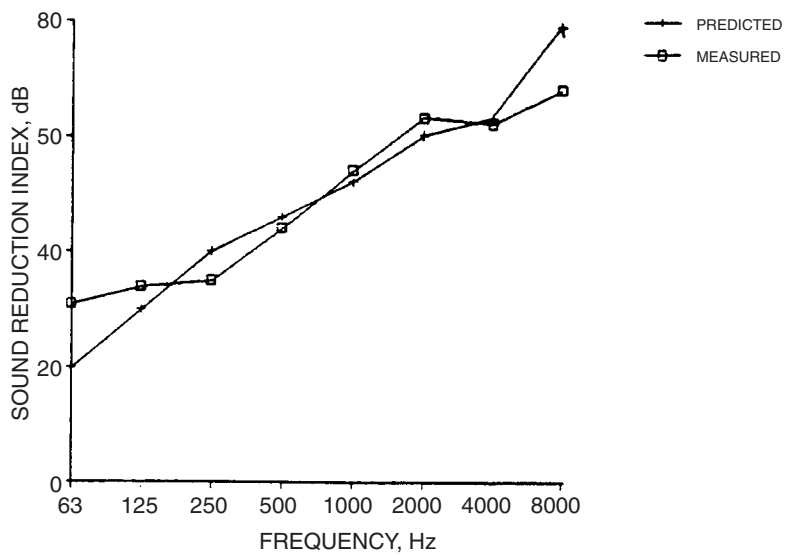


FIG. A-25 Predicted and measured sound reduction index of panel 2. (Source: Altair Filters International Limited.)

Experience thus far. It has been shown that the acoustic performance of lined and unlined panels can be predicted with reasonable accuracy for flat and corrugated panels. It has also been shown that, where noise control is important, unlined corrugated panels are not recommended unless other engineering considerations dictate their use, because corrugated panels are intrinsically less effective as sound insulators than flat panels of the same thickness.

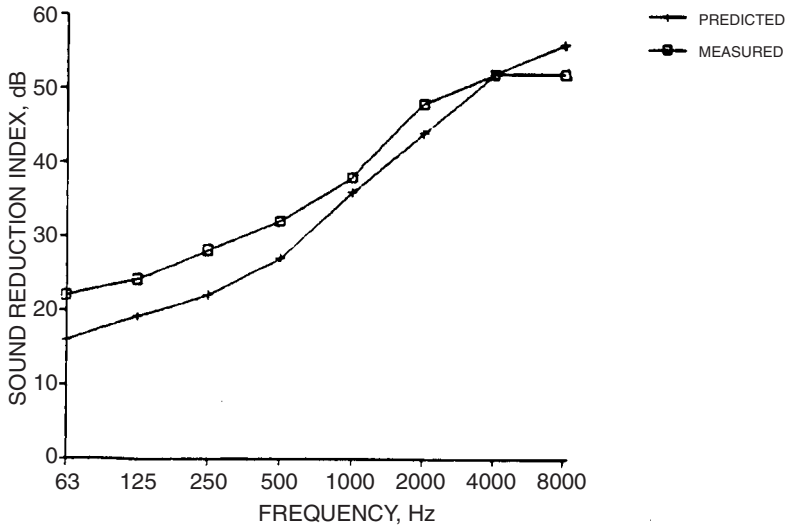


FIG. A-26 Predicted and measured sound reduction index of panel 3. (Source: Altair Filters International Limited.)

A lining of sound absorptive material can substantially increase the sound reduction index of panels and the additional attenuation depends on the density, fiber diameter, and thickness of the lining. By careful selection of these parameters, the acoustic disadvantages of corrugated panels can be considerably reduced so that corrugated panels can be used confidently in situations where noise control is a primary requirement. The additional bending stiffness of corrugated panels permits a thinner outer skin to be employed and reduces the amount of additional bracing required to provide the structural integrity necessary in the demanding environment offshore. This reduction in overall weight compensates for the additional material used in forming the corrugations.

By careful design of the panel, a corrugation profile can be selected, which provided the most cost-effective solution when structural integrity, weight cost, ease of manufacture, and acoustic performance are considered. When expensive materials, such as stainless steel and aluminum, are employed, the reduction in cost by using a thinner-walled corrugated panel can be considerable.

A further consideration is the fire rating of lined corrugated panels. The normal requirement for bulkheads and decks offshore is the "A-60" class division. Corrugated panel designs of the type described here have been submitted to, and approved by, the appropriate authorities.

In some situations where a particularly high acoustic performance is called for, the corrugated design lends itself well to a multilayer construction employing an additional inner layer of heavy impervious material. Cheaper materials are used for the additional septum rather than for the outer skin. The acoustic attenuation of these multilayer designs is comparable to the performance of flat panels employing outer skins of twice the thickness of the corrugated outer skin. Figure A-27 compares the measured performances of a traditional 5-mm-thick flat panel design with a 100-mm-thick absorptive lining and a multilayered panel based on a 2.5-mm-thick corrugated panel lined with a 50-mm absorptive layer.

The nominal surface weights of the two designs are 50 kg/m^2 and 40 kg/m^2 for the flat and corrugated panels, respectively. Except at 63 and 125 Hz, the performance of the two panels is very similar.

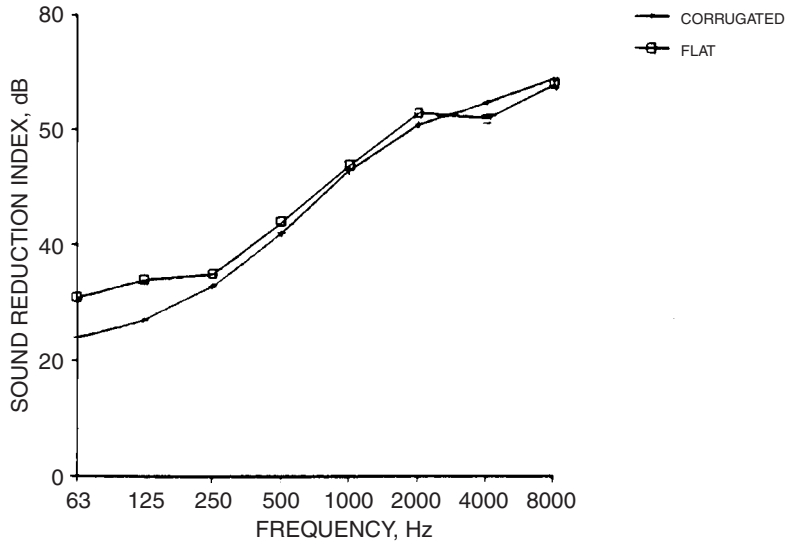


FIG. A-27 Predicted sound reduction indices of two high-performance panels. (Source: Altair Filters International Limited.)

In summary. The acoustic performance of corrugated and flat steel panels can be predicted. The acoustic behavior of corrugated panels is very different from that of flat panels. This means that if corrugated panels are required, careful consideration must be given to the design, since unlined corrugated panels are unsuitable on their own for noise control applications.

However, the greater bending stiffness of corrugated panels offers many financial and structural advantages in the demanding environment that exists offshore, especially for gas turbines.

By lining the interior of a corrugated panel with a material whose physical parameters have been carefully chosen, the inherent acoustic weaknesses can be overcome. Thus a more cost-effective approach to gas turbine enclosure design can be adopted, which considers the structural integrity, weight, cost, ease of manufacture, and acoustic performance. The resultant designs employ less bracing and thinner outer skins to achieve the same acoustic performance as flat-walled constructions weighing typically 25 percent more than the equivalent corrugated design.

Actuators

Actuators, Electrohydraulic

Electrohydraulic actuators are among the more common varieties of actuators in the process plant market and are also more accurate in terms of position control. These components have very specific (to a particular manufacturer) design components. Therefore, terminology in the detailed descriptions that follow is specific to the information source, J.M. Voith GmbH in this case. In the case of requesting competitive bids, the end user should consider requesting similar or alternate features.

Areas of Applications Benefits*

- Reliable and highly accurate conversion of electrical control signals into specific process values.
- Combination of electronics, sensor technology, and mechanics resulting in reduction of interfaces and high degree of reliability.
- High regulated magnetic forces (use of Hall effect) make it possible to apply robust magnetic drives.
- No need for external regulating equipment since complete regulating system is integrated in the chassis of the control unit for the magnetic drive (high degree of EMV resistance).
- Parameters for controller output range can be set from outside.
- High degree of reliability.
- Infinitely variable conversion of input signal i_E into output modes power, pressure, or stroke with high dynamic force.
- Inversion range $E \Leftarrow 0.05\%$.
- Conversion time for 50% regulating value 25 msec.
- Integrated sensor technology and control electronics with function monitoring and actual value remote display output in robust housing or in pressure-resistant casing.
- Electrohydraulic alternative to the retrofitting and modernization of mechanical/hydraulic control and regulatory systems.

Figures A-28 to A-37 and their descriptions outline a typical comprehensive range of electrohydraulic actuators. Different designs may be designated with a specific trademark. This is indicated where relevant.

Aerfoils; Airfoils (*see* Metallurgy; Turbines)

Agitators

Broadly speaking, agitators can be used to produce the following:

1. Uniformity between different components, solid or liquid, miscible, or otherwise. This produces liquid blends or solid suspensions (*see* Gravity Blending in the section on Tanks).
2. Heat or mass transfer between matter. Applications include extraction and leaching processes (*see* Oil Sands).
3. Phase changes in a mixture. Homogenizing, emulsification, and crystallization are among these processes (*see* Centrifuges).

Reference and Additional Reading

1. Bloch, H., and Soares, C. M., *Process Plant Machinery*, 2d ed., Butterworth-Heinemann, 1998.

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* Source: J.M. Voith GmbH, Germany. Adapted with permission.



Power station applications:

Controlling the power output of gas and steam turbines.



Paper mills:

Controlling process demands steam pressure and grid power.



Chemical and petrochemical plants:

Controlling energy and mass flows.



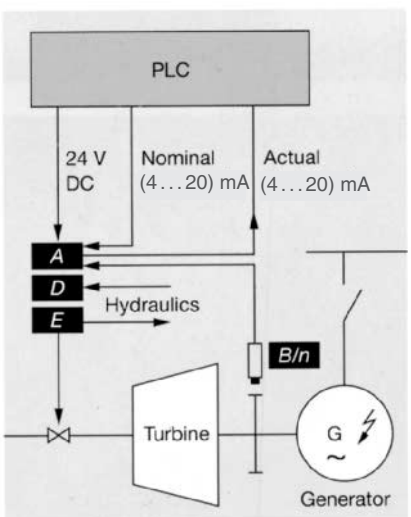
Food processing plants:

Controlling energy flows.

FIG. A-28 Applications of hydraulic actuators by industry and control function. (Source: J.M. Voith GmbH.)

Proportional and dynamic conversion of electrical control signals (0/4 . . . 20) mA into power, regulating pressure, regulating stroke and rpm is achieved with highly versatile modular component technology:

- **A** Control unit: magnetic drive with integrated control electronics in standard and Ex-protection housings.
- **B** Sensor technology: stroke **(B/s)** and rpm sensors **(B/n).**
- **C** Servo valve to control discharge quantities.
- **D** Pressure reduction valves.
- **E** Various operating cylinders.



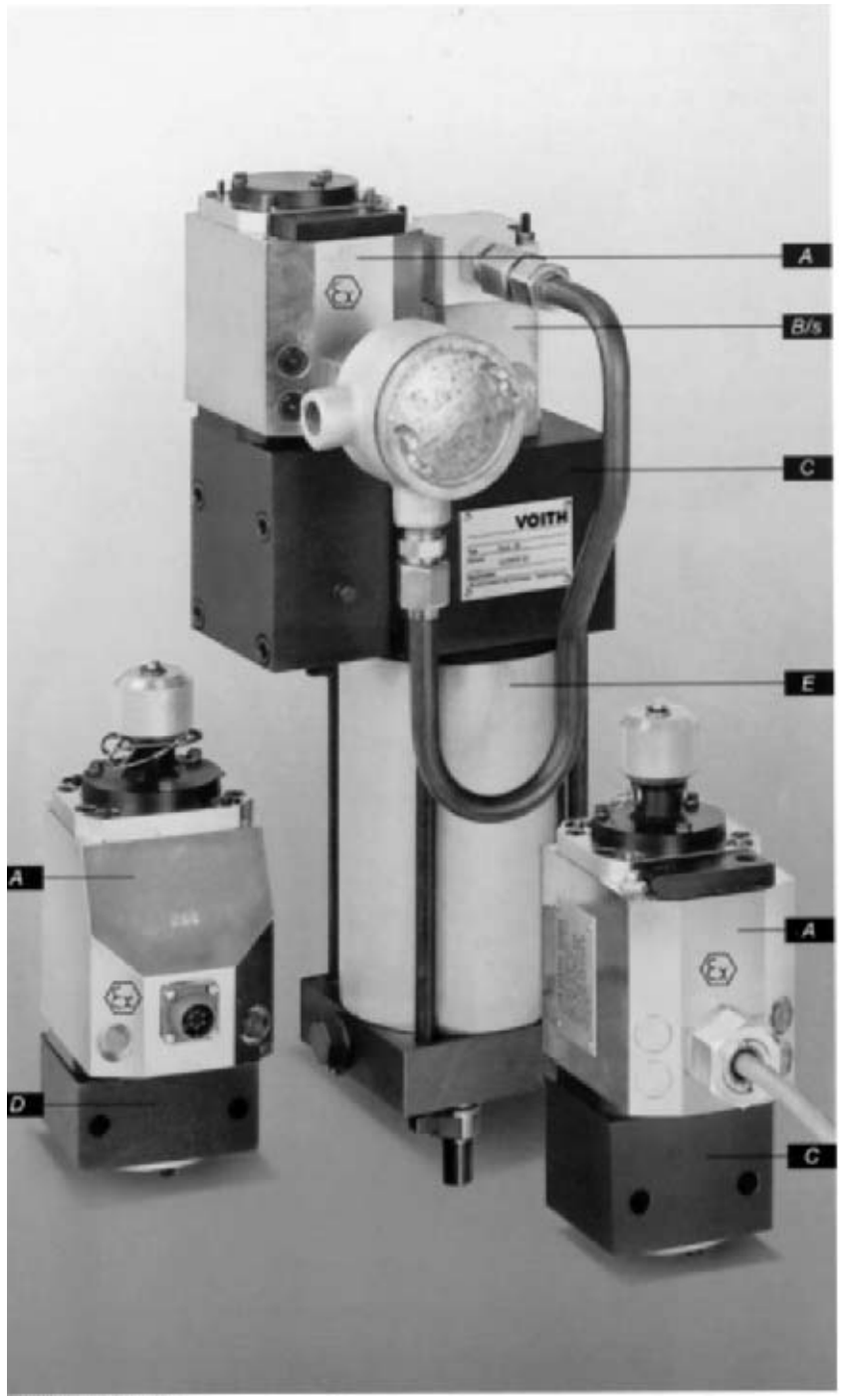


FIG. A-29 Modular component units used for conversion of electrical control signals. (Source: J.M. Voith GmbH.)

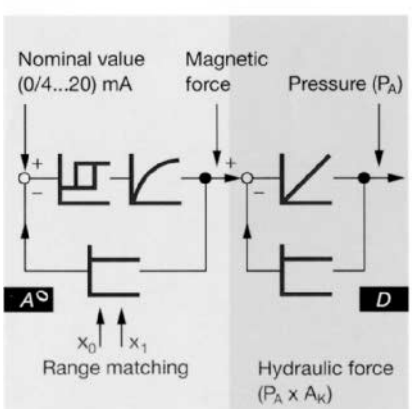
Basic item of the modular assembly is a power-regulated electromagnet consisting of the following components: magnet housing, coil, anchor with driving rod, and the control electronics.

A regulator is used to keep the degree of linear force applied to the anchor at a rating proportional to the input signal.

This ensures that the output (force depending on pressure output) is not affected by adverse influences such as hysteresis, air gap etc (DBP-2019345).

Forces are achieved that are applied directly via the driving rod, i.e., without any hydraulic amplification, to a control piston or a stroke valve.

The diagram below shows how the neutral position magnetic force on one side of the piston is checked by a force proportional to the reduced pressure P_A. The result is a direct and constant match between the input signal and the reduced pressure.



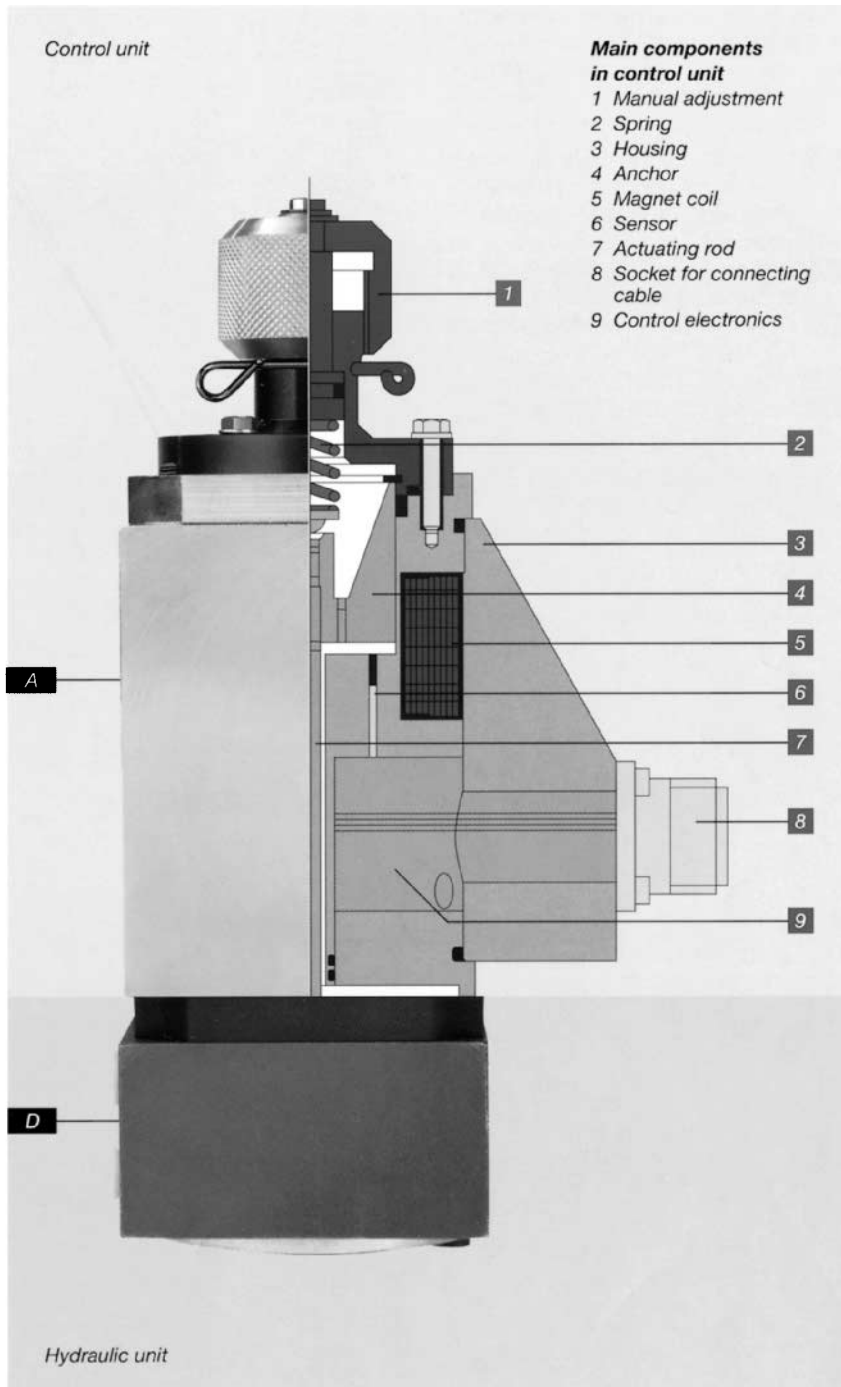


FIG. A-30 How actuators function: power-regulated electromagnet. (Source: J.M. Voith GmbH.)

The section diagram illustrates the main features of control regulator functioning:

- Drive and control pistons with failsafe spring return.
- Internal oil circulation as part of closedown process (rapid closedown ≤ 0.1 sec).
- Inductive stroke pick-up (7) with clamp magnet coupling (8).
- 400 N magnet drive (1) with integrated control electronics for control pistons (3) and position of piston rod (14)

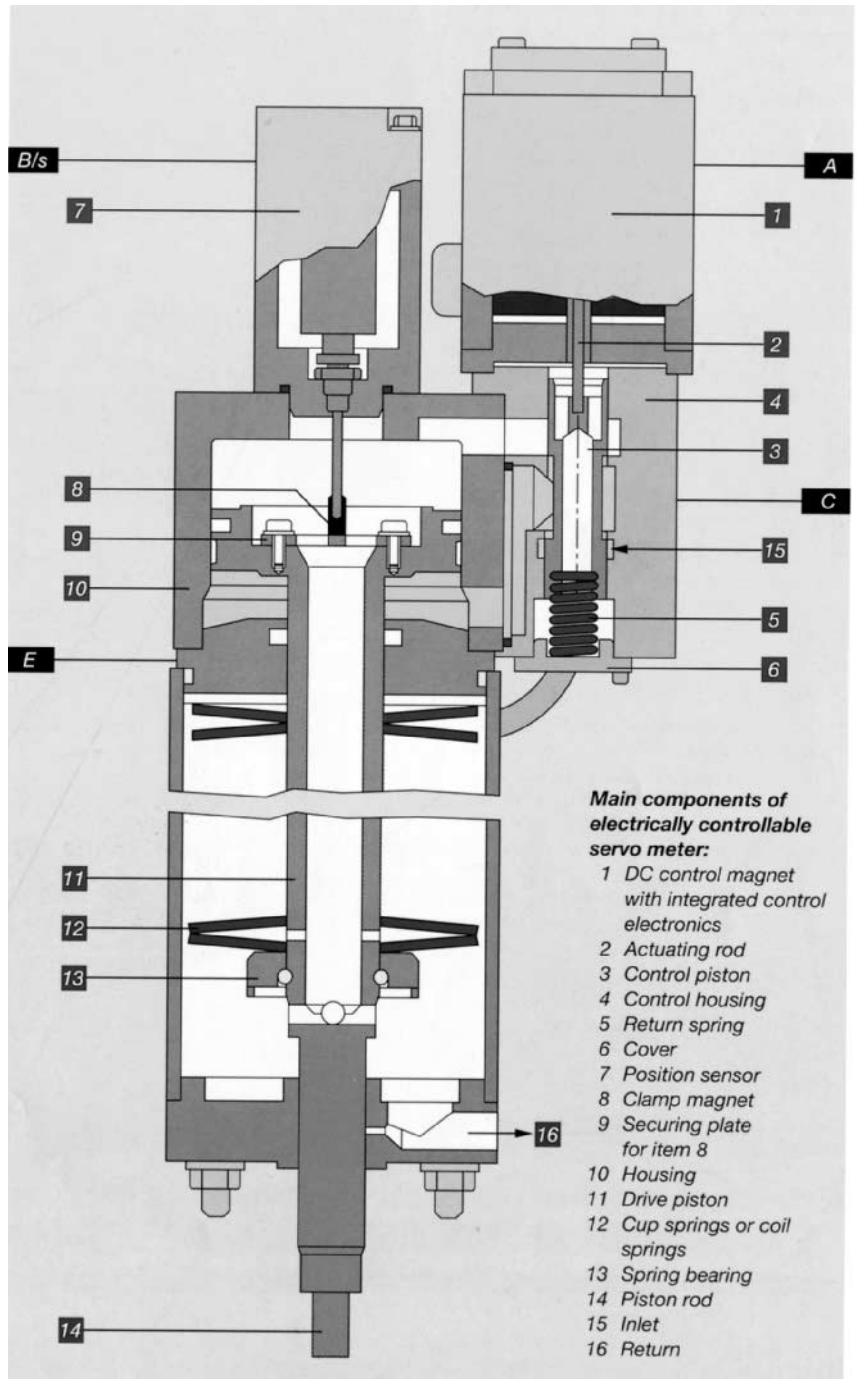


FIG. A-31 Control regulator functioning and main features. (Source: J.M. Voith GmbH.)

Electrically controllable pressure regulating valve (with or without PID controller)

consisting of A and D.
Converts (0/4…20) mA in (0/7); (0/16); (0/60) bar (Δ Q ~ 50 $\frac{ltr.}{min.}$)

A directly applied controlled magnetic force F is brought into exact counterbalance with a proportional hydraulic force. The appropriate output pressure in relation to the input signal i_E is controlled by X_0 and X_1. Conversionis effected with a loss of < = 0.1%.

In the version incorporating a PID controller you can compensate for pipeline pressure losses or you can control valve positions and turbine rpm.

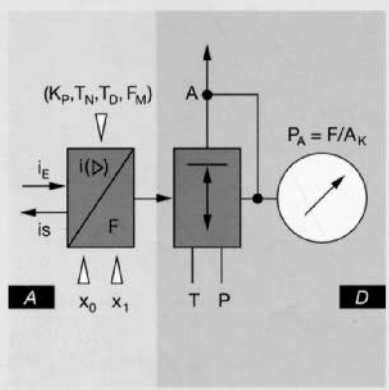


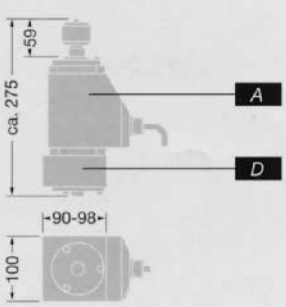
Parameters without integrated PID controller

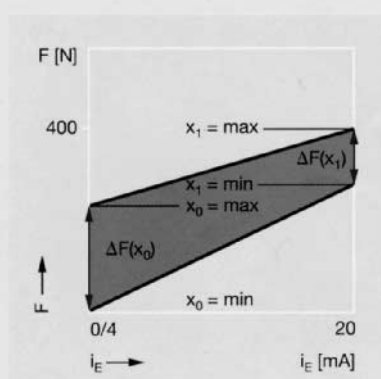
X_0 = *pressure p_A at i_E = 0 or 4 mA*
X_1 = *pressure p_A at i_E = 20 mA*

Parameters with integrated PID controller

(e.g. for position regulation)
K_P = *proportional amplification*
T_N = *integral action time*
T_D = *deviractive action*
F_M = *gate valve centre*
x_0 = *position of motor at i_E min*
x_1 = *position of motor at i_E max*
is = *remote display (position)*







Directly controlled gate valve (with or without PID controller)

consisting of A and C.
Controls passage cross-sections (flow control):
(12-20) mA ≙ (0-200) mm² inlet
(12- 0) mA ≙ (0-700) mm² return

With gate valves a controlled magnetic force F is brought into counterbalance with an elastic force, i.e., a dependent force. The input signal i_E is allocated the appropriate cross-section for the valve with X_0 and X_1. The decisive feature is hysteresis-free control in the area around the dydraulic middle position. Symmetrical or asymmetrical controlled cross-sections A can be controlled directly up to 700 mm². The Turcon® CTo version with protection against over-speed rpm is available as a specially adapted gate valve.

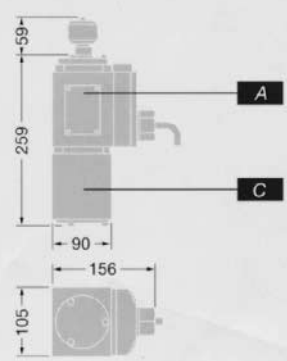


FIG. A-32 Control regulator valves. (Source: J.M. Voith GmbH.)



Electrically controlled regulator

consisting of A, B, C, E.

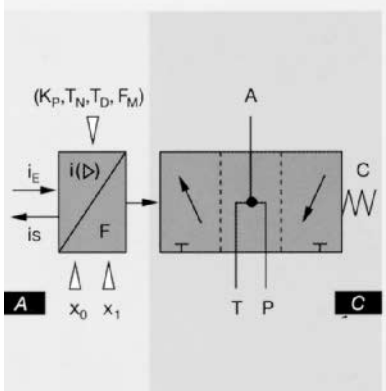
In principle the way that regulators are controlled is via a gate valve for which the magnetic drive has both a magnetic force controller and a superimposed position regulator. The set size of the position regulator acts as the reference value for force control. The input signal i_E—in this example the reference value for regulation—is allocated via X_0 and X_1 the appropriate stroke from the drive piston which is displayed by a (4...20) mA signal. If the control deviation is excessive this is displayedvia a potential-free optocoupler output. In order to linearize flow lines on flaps and valves the control electronics can be enlarged by the addition of a 10-stage function indicator.

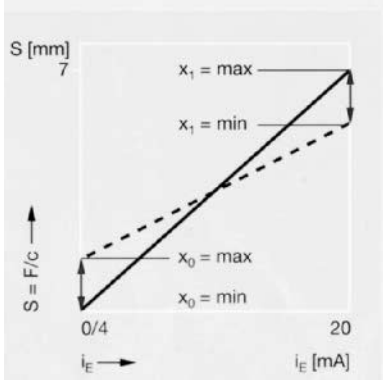


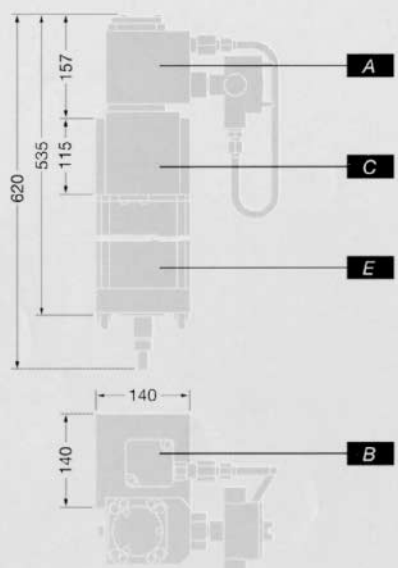
Special technical data

Operating pressure:	6…100 [bar]
Working stroke:	5…60 [mm]
Controlled forces:	500…14000 [N]
Control response times (min):	100 [m/sec]
Positioning accuracy:	15 [µm]

Special models made to order.







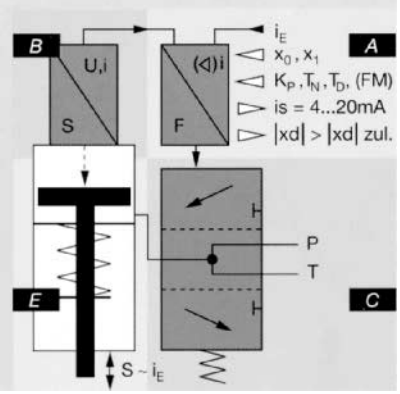


FIG. A-33 Electrically controlled regulator. (Source: J.M. Voith GmbH.)

A

Magnetic drives with integrated control electronics

Input: Signal current (0/4...20) mA

Output: Magnetic force: Initial value of X_0 adjustable from (0 to 250) N. Final value of X_1 adjustable from initial value up to 400 N.

Hysteresis: ≤ 0.05%

Dynamics: Corresponds to a PT2 component with τ = 15 msec and D = 0.8

Versions available: with and without manual adjustment, with and without Ex-protection, and with and without integrated PID controller for additional, dynamically demanding control and regulation tasks (e.g., position or rpm regulation.)

B/s

Sensor with integrated evaluation/read-out electronics

For strokes of (0 to 40) and (0 to 60) mm in pressure-resistant enclosed casing as per EEx d with short cable lead and standard version supplied with 3-pole plug.

A + D

Magnetic drive + pressure reducing valve = pressure regulating valve

This combination converts (0/4 . . . 20) mA into (0 to 7), (0 to 16) or (0 to 60) bar. Respective control piston diameter readings are: 26, 18 and 10 mm.

A + C

Magnetic drive + control valve (3/3 or 4/3) = gate valve

The control valve is at the hydraulic centre (adjustable using magnetic force) at about 12 mA. An increase to 20 mA moves the valve against a spring force to the inlet fully open position (approx. 200 mm^2). A decrease to (0/4) mA brings it to the return fully open position (approx. 700 mm^2).

A + B + C + E

Magnetic drive with PID controller + position sensor + control valve + regulating cylinder with inbuilt return spring = regulator

For converting (0/4 . . . 20) mA into controlled strokes of (0 to 30), (0 to 60) mm. Flow forces in open directionca. 15000 N. Spring forces in closedirection ca. 9000 N. Time taken from open to close ≤ 0.10 sec.

FIG. A-34a Drive/control valve options. (Source: J.M. Voith GmbH.)

Technical data for all electronic component assemblies

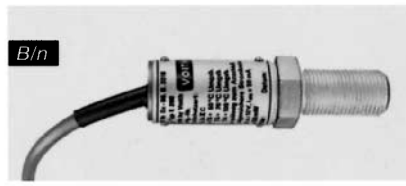
Operating voltage U_B	(20 ≤ U_B ≤ 30) V
Operating current i_B	approx. 0.5 A (up to 2.4 A when switching-on)
Surface temperature T	(-40 ≤ T ≤ 85)°C
Storage temperature T_L	(-60 ≤ T_L ≤ 150)°C
Type of protection	DIN 40050 IP 65
Protection against explosion as per PTB-No. EX-90.C.1065	EEx d IIC T4 (T5)*
EX-connection	Short cable lead or EExe distributor
Standard connection	Mil-plug (6 pole)
Impact resistance	18 impulses of 50 g over 0.005 secs.
Vibration	18 to 500 Hz at 4.2g
Humidity	100% condensing

* as per US standard currently being prepared.

FIG. A-34b Technical data for electronic component assemblies. (Source: J.M. Voith GmbH.)

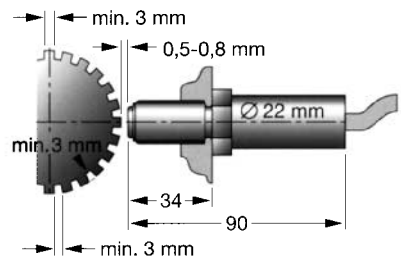
Sensors that are encased in robust EEx d housings operate more reliably when used for constant measurement of the rpm and valve positions in compressors and gas and steam turbines. This type of housing protects them from adverse environmental factors (EMV, temperature changes, humidity, and oscillations).

RPM sensors



Technical data Type E 288
Draw-No. 43.9423. xx

Measurement cog module:	≥ 2.5
Cog shape:	any – square profile preferred
Distance between sensor and cog:	0.5 . . . 0.8 mm
Measurement frequency:	25 Hz . . . 15 KHz
Operating temperature:	–20 . . . +125°C
Type of protection as per DIN 40050:	IP 65
Protection against explosion:	EEx ib IIC T4 . . . T6



Technical data of sensors 90.1265.xx and 90.3131.xx

As above, but without the impulse ammplifier and without Ex-protection*.These sensors have a coil resistanceof ca. 1.1 kΩ at 25°C and areauthorized to operate at temperaturesof up to 150°C.

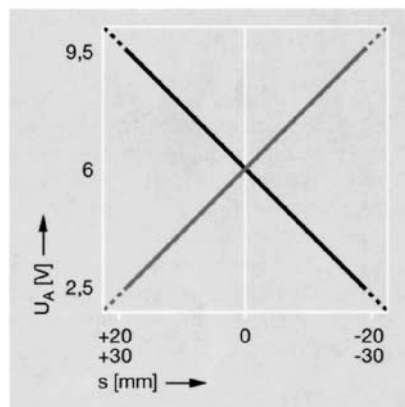


90.1265.xx 90.3131.xx

*) *EEx d IIC T5 – version available shortly.*

Position sensors





Technical data

Supply voltage:	18 . . . 30 V/DC
Measurement range:	Type 40 ± 20 mm Type 60 ± 30 mm
Output voltage for rising or falling trace:	see diagram
Power consumption max.:	50 mA
Pressure resistance (static):	350 bar
Type of protection as per DIN 40050:	IP 65
Protection against explosion:	EEx d IIC T5

FIG. A-35 How sensor technology works. (Source: J.M. Voith GmbH.)

Turcon® CTo is an electrohydraulic rapid trigger shut-off device being a special version of the gate valve. Turcon® CTo is used mainly as a monitoring system resp. as an rpm protection device for high-rpm steam turbines. It offers major advantages compared with the mechanical-hydraulic systems available hitherto:

- Very high degree of reliability as a result of constant trigger speeds independent of oil temperature, electromagnetic direct drive with failsafe return/reset. The rapid trigger device can be tested without interrupting operations by means of 10 msec test impulse using trigger frequency generator (integrated in magnetic system).
- Easy to install and service: the magnetic valve and electronics are all one unit. Alignment takes place at the factory, removing the need for labor-intensive adjustments during commissioning.
- High reliability as a result of robust construction and low pressure hydraulics.
- Contact-free speed readings, avoiding strain on main bearings from load displacement in case of overspeed rpm. This makes the device suitable for turbines with extremely high rpm ratings.
- discontinuity in wire and/or short circuit at speed sensor.
- Housing temperature $\geq 85°C$.
- Supply voltage ... $\geq$ (30 ... 33) V/DC.
- Irregular loss of speed -tripping speed (n_{trip}) $\geq (1.03 \cdot n_{trip})$ (10 msec-Test) $\geq (n_{trip} \pm 0.5\%)$.
- External switchoff (activation of standby power circuit).

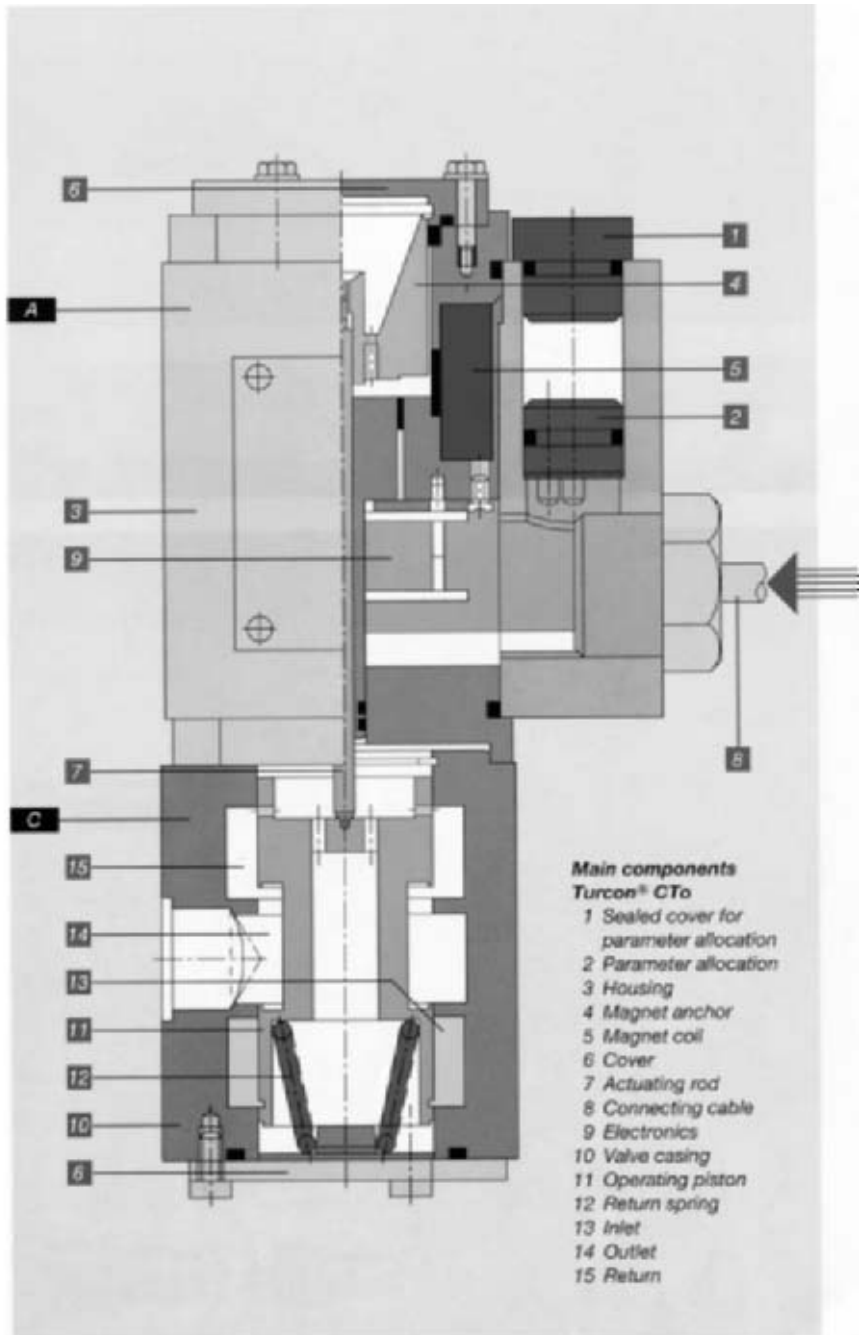


FIG. A-36 Speed protection device operation. (Source: J.M. Voith GmbH.)

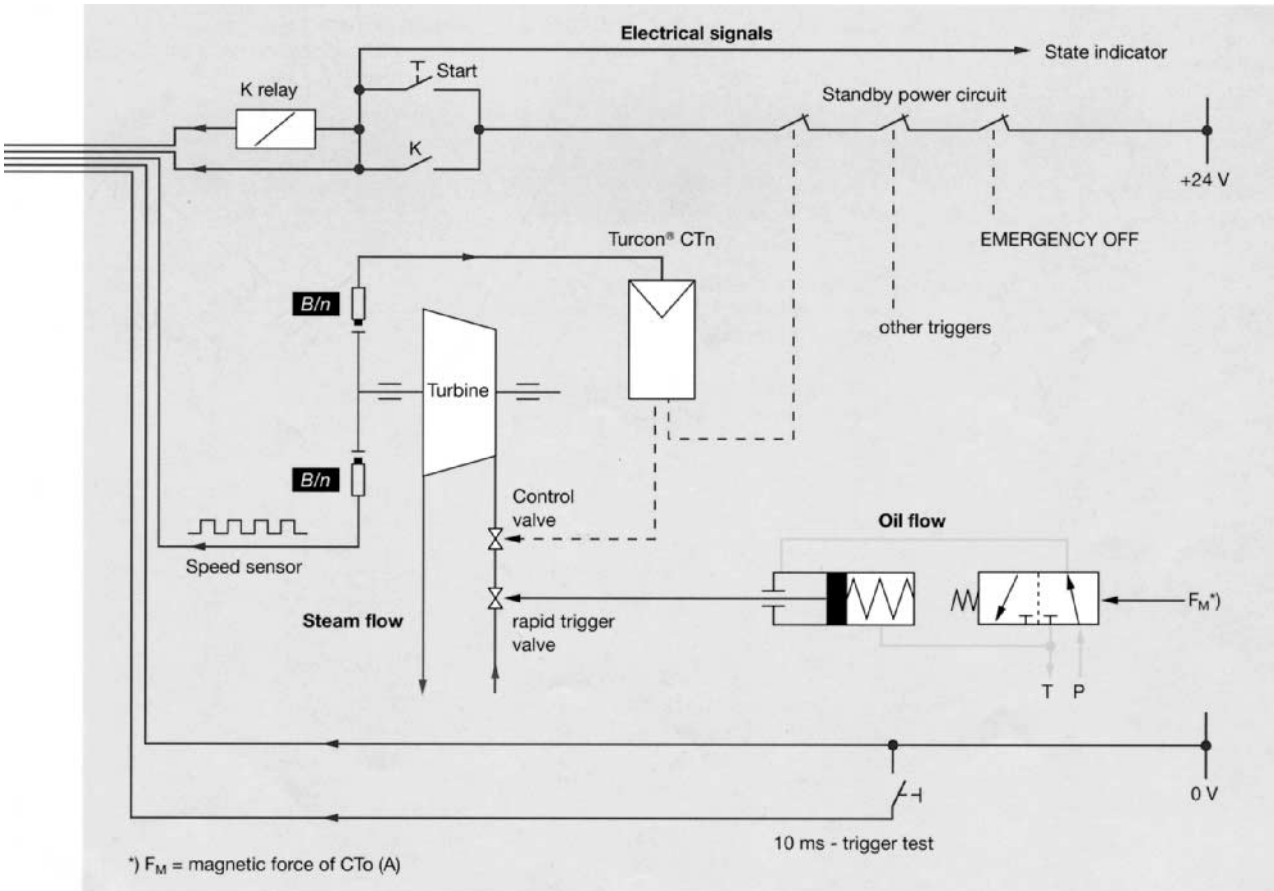


FIG. A-37 Trigger criteria for protection against overspeed. (Source: J.M. Voith GmbH.)

Agriculture; Agricultural Product Processing

(*See also* Ecological Parks; Environmental Accountability; Forest Products; *other related topics*.)

Agriculture is too wide a field to be dealt with comprehensively in this book. However, many generic types of equipment used in this field are discussed, including centrifuges, conveyors, pumps, motors, chillers, and so forth. Options, such as specific material selections, for instance, plastic gears and lobes (instead of metal ones) in pumps handling food, may alter overall designs. Agricultural product machinery is often custom designed or has customized options for this reason. Certain machinery types most commonly used in, if not unique to, the agricultural industry have been essentially left out of this book. These types include: pelletizers (such as might be used for making food pellets), briquette makers, and homogenizers (for milk for instance).

Although agricultural machinery might be simpler than, say, machinery used in a modern plastics plant, there is a growing sophistication with all forms of the process industry, such as coolers in agriculture. *See* Cooling.

Agricultural products that support the industry are similarly frequently custom designed or specified, such as the insecticides made for agricultural crops. There is no uniformity in quantities used for application either. For instance, 80% of all the agricultural insecticides used on crops in the United States are used on the cotton plant.

One indicator targeted for optimization, for the process engineer handling agricultural products, is reduced chemical pollutants that originate from a process. The potential for decreasing chemical pollutant levels in product handling increases with technical developments. Methods for reducing these levels are frequently provided by biological engineering means, including:

- Bioremediation of polluted soil
- Use of naturally occurring pesticides instead of chemical pesticides
- Breeding plants/crops with characteristics that enhance production without further chemical use

Examples of such technology include the ability to develop grazing grass and crops with an aerated root system that will resist drought, floods, and also potentially neutralize toxic mineral compounds by oxidizing them.

Firms in the agricultural industry are excellent candidates for joining ecological industrial parks. They must have the highest standards of cleanliness and have a great deal to offer a group of industries in terms of experience in this area. If a metal workshop and industrial furnace can coexist in the proximity of a milk homogenizing facility, health conditions for all will improve and pollutants, overall, will drop. If agricultural firms can thus convey the environmental practices they must abide by, the industry as a whole, and the conditions under which they must work, will automatically improve.

References and Additional Reading

1. Soares, C. M. *Environmental Technology and Economics: Sustainable Development in Industry*, Butterworth-Heinemann, 1999.
2. Comis, D., "Miracle Plants Withstand Flood and Drought," *The World and I*, February 1998.

Air Filtration; Air Inlet Filtration for Gas Turbines

One of the most common applications of air filtration in a process engineer's world is filters at the air intake of a gas turbine. These filters take a toll on the gas turbine's thermal efficiency and therefore increase the turbine's fuel consumption, so their designers make every attempt to minimize pressure drop across the filter elements.

Industries and applications where these filters are used include refineries and chemical plants, the food industry, compressors, power stations, electrical generators, warehouse and building air-conditioning systems, as well as computers and electrical cabinets.

Purposes for installing gas turbine air-inlet filtration include

- Prevention or protection against icing
- Reduction/elimination of ingestion of insects, sand, oil fumes, and other atmospheric pollutants

Potential ice-ingestion problems can be avoided with a pulse-jet–type filter, commonly called a *huff and puff* design. Ice builds up on individual filter elements

that are part of the overall filter. When a predetermined pressure drop is reached across the elements, a charge of air is directed through the filter elements and against the gas turbine intake flow direction. The ice (or dust "cake") then falls off and starts to build up once more.

Ice ingestion has caused disastrous failures on gas turbines. A few companies also make instrumentation that will detect incipient ice formation by measurement of physical parameters at the turbine air inlet. Sometimes, if the pulse-type filter is retrofitted, the anti-icing detection instrumentation may already be there. The pulse filter, however, provides a preventive "cure" that will work, regardless of whether the icing-detection instrumentation is accurate, as the cleaning pulse is triggered by a signal that depends on differential pressure drop across the filter elements. If a pulse filter is used, icing-detection instrumentation, which is normally necessary in a system that directs hot compressor air (bleed air) into the inlet airstream, is not required.

Many filtration applications are examples of retrofit engineering or reengineering because the original application may have been designed and commissioned without filters or the original choice of filters/filter elements was inappropriate. For tropical applications, filter media that swells or degrades (also rain must not be allowed to enter the filter system) cannot be used because of the intense humidity. This will exclude cellulose media. Tropical installations present among the most severe applications.

Inlet Air Filters for the Tropical Environment*

The factors that determine design include the following:

Rainfall

The tropics extend for 23°28′ either side of the equator stretching from the tropic of Cancer in the north to the tropic of Capricorn in the south and represent the tilt of the earth's axis relative to the path around the sun. The sun will pass overhead twice in a year, passing the equator on June 21 on its travel north and September 23 on its travel south. The sun's rays will pass perpendicular to the earth's atmosphere and so will have the least amount of filtration, giving high levels of ultraviolet rays.

The area has little seasonal variation; however, the main characteristic of the area is the pronounced periods of rainfall. Typhoons and cyclones are common to certain parts of this area.

It is not surprising that the records for the highest rainfall ever recorded are all within the tropics. Intense rainfall is difficult to measure since its maximum intensity only lasts for a few minutes. Rainfall can be expressed in many ways, either as the precipitation that has fallen within 1 hour (in millimeters per) or over shorter or longer periods but all relating back to that same unit of measurement. Since gas turbines experience problems due to rainfall within a few minutes, it is important to take account of the values of "instantaneous rainfall" that can occur.

The most intense rainfall ever recorded was in Barst, Guadeloupe (latitude 16°N), on November 26, 1970, when 38.1 mm fell in just 1 min.

Another important feature regarding rainfall is the effect of wind speed. "Horizontal rain" is often described, but in practice is unlikely to occur. However, wind speeds can give rain droplets significant horizontal components. The impact

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* Source: Altair Filters International Limited, UK. Adapted with permission.

of this can be very important, particularly with small droplet sizes. Even droplets 5 mm in diameter will cause a vertical surface to be almost 4 times wetter in wind speeds of 37 m/s than on the horizontal surface (to which the rainfall rates relate).

The effect of rainfall in tropical environments on the operation of gas turbines has been very much underrated.

The humid environment also ensures that relative humidities are generally high, with the lowest humidities being experienced during the hottest part of the day and the highest occurring at night. During the rainy season, the humidity tends to remain constant throughout the day. The effect of humidity is important where airborne salt is concerned since salt can become dry if the humidity is below 70%.

Dust

Dust levels in tropical environments in southeast Asia are generally low.

There are, of course, always specific exceptions to this, for example, near new construction sites or by unpaved roads. But, in general, dust is not a significant problem. Mother nature ensures this by casting her seeds on the fertile soil and quickly turning any unused open space into a mass of overgrown vegetation very quickly, thereby suppressing the dust in the most natural way.

Insects and moths

In the tropics the hot, humid environment is a natural encouragement to growth of all kinds. It is often said that if a walking stick is stuck into the rich fertile soil of the area and left for 3 months, it will sprout leaves and grow. Certainly the insect population reflects this both in size and quantity. Large moths are common to the area and tend to occur in quantity during specific breeding periods. These can quickly cover intake grills, obstructing airflow and even causing large gas turbines to trip. Some of the largest moths are found in the tropics. A common moth in India and southeast Asia is the Swift moth (Hepralidae), which can have a wing span of some 15 cm and is said to lay up to 1200 eggs in one night. Another moth is the Homoprera shown in Fig. A-38, which has a similar wing span. Moths are attracted by the lights that often surround the turbine installations, as well as the airflow, which acts as a great vacuum cleaner.

On one installation in Sumatra, large gas turbines have been known to trip out after only 8 hours of operation due to blockage of the air filters with moths.

Fortunately, moths tend to confine themselves to within a few miles of land and so offshore installations do not tend to suffer these problems.

Problems Experienced

Many feel that standardization is the key to reducing costs and boosting profits. It is not surprising, therefore, that gas turbine air-filter systems were designed with this in mind.

Dust was important to system designers, and so filter systems may be chosen to be able to deal with prodigious amounts of it, whereas, in practice, dust is only normally a problem next to unpaved roads or construction sites.

Despite this, most gas turbines were fitted with elaborate and expensive solutions to overcome a problem that hardly existed, or at least only in a relatively small percentage of installations. Many of these systems employ bleed fans that need additional electrical energy and a constant maintenance requirement. A typical system is shown in Fig. A-39. This system employs spin tubes that swirl dust to the outside of the tube where a bleed slot extracts the dust while allowing the

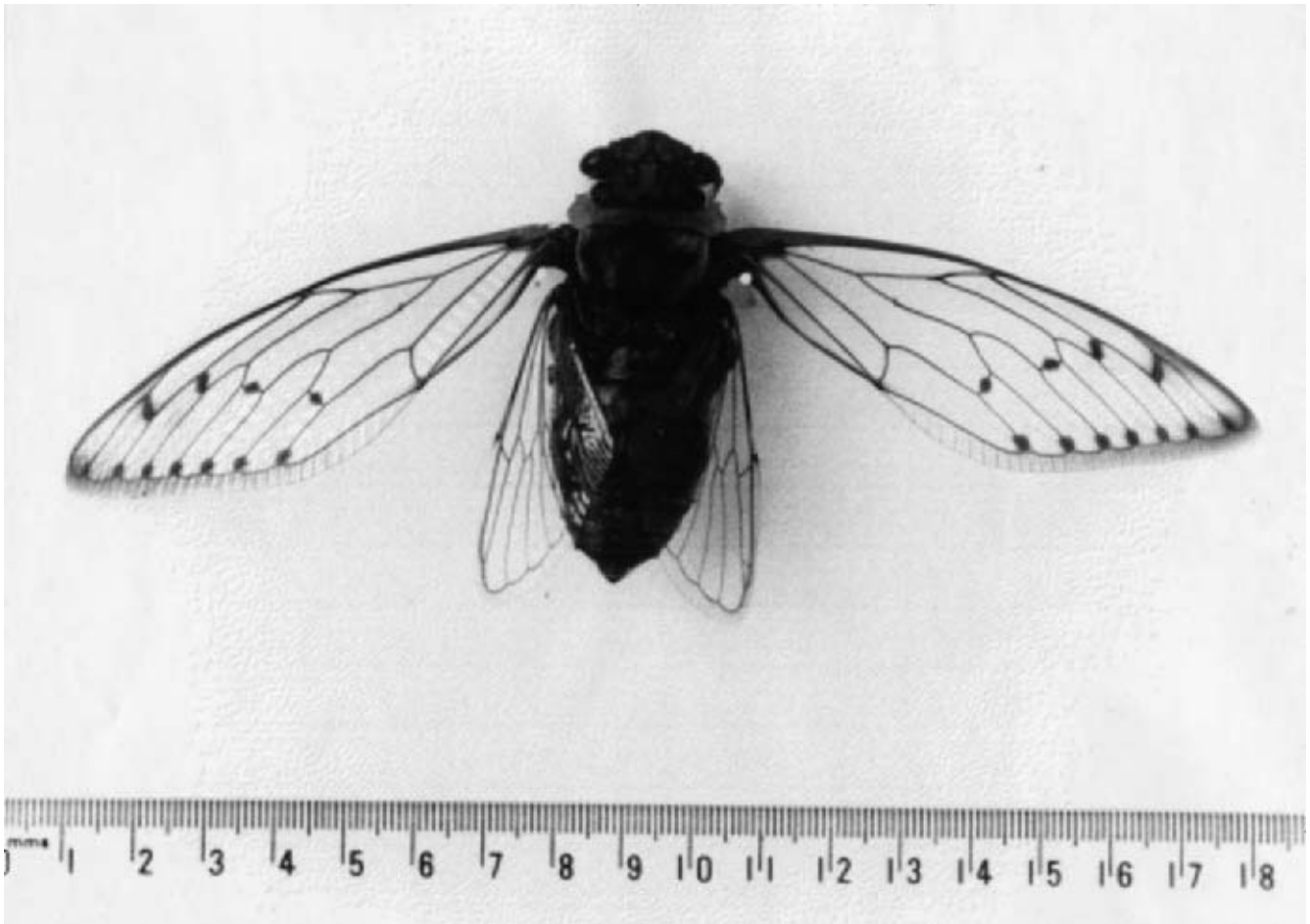


FIG. A-38 Homoprera (insect type). (Source: Altair Filters International Limited.)

cleaner air to pass through the main core of the tube. Since the air is rotated, the peripheral speeds need to be high, which, in turn, results in a relatively high pressure-loss coefficient. The efficiency of the system is very reliant on the bleed air, which is provided by auxiliary fans.

Another bleed extract system uses a series of convergent vanes that funnel the air toward a central slot through which bleed air is extracted. The heavier dust particles are guided toward the bleed extract, while the main air passes between the vanes at almost 180° to the general direction of airflow. Again, the efficiency of the system is reliant on the provision of bleed air.

Protection against rain was elementary. On many systems that had dust-extract systems, no further provision was made. On others a coarse weather louvre, often of plastic, was provided, as shown in Fig. A-40. Sometimes a partial weatherhood was provided, sometimes not.

The emphasis on the designer was to provide a "three-stage" filter system, without worrying too much about the suitability of those stages.

There was recognition of the high humidities that exist, and so most systems incorporated a coalescer, whose function was to coalesce small aerosol droplets into larger ones, which could then be drained away. These coalescer panels varied

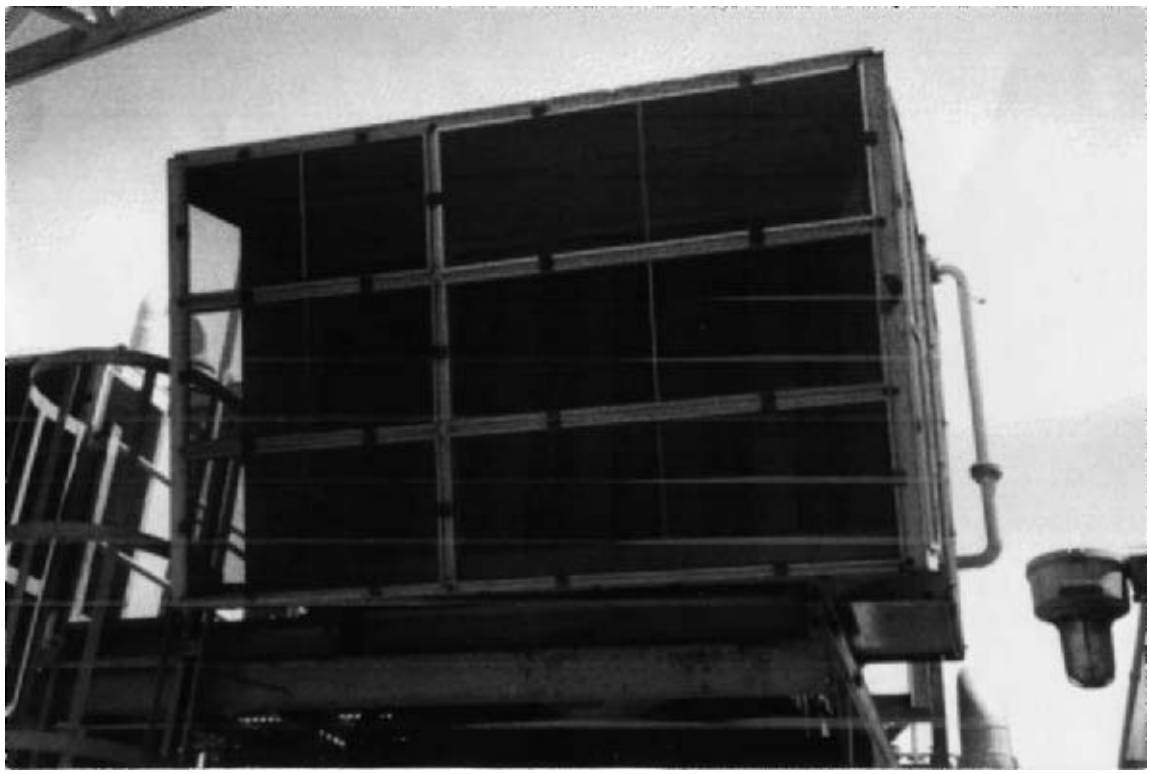


FIG. A-39 A typical spin-tube inertia filter. (Source: Altair Filters International Limited.)

between pieces of knitted mesh wound between bars within the filter housing to separate panels with their own framing. Loose glass fiber pads were used as an inexpensive solution and also served as a prefilter pad.

The final stage in almost all of these systems was the high-efficiency filter element, either as a cartridge or as a bag. The high-efficiency cartridge was typically a deep-pleated glass-fiber paper sealed in its own frame and with a seal on its rear face. A typical example is shown in Fig. A-41. Other high-efficiency filters employed glass-fiber pockets that fitted into permanent wire baskets within the filter house.

Almost all of the filter systems were enclosed in housings constructed in carbon steel, finished with a variety of paint finishes. Other materials, such as stainless steel, were not common since their initial cost was thought to be excessive.

Protection against complete filter blockage was often provided by means of a bypass door. This normally was a counterbalanced door in which a weight held the door closed. When the pressure drop across the filter was high, this overcame the force exerted by the balance weight and the door opened, thereby bypassing the filter system with unfiltered air. A typical system is shown on the top of the filter housing shown in Fig. A-40.

Operational Experience

The ultimate test of the filter system is whether it is providing the required protection to the gas turbine. Unfortunately, there are mainly instances where problems have been found. The following figures illustrate problems that have arisen with this information source's designs, as installed in the field:

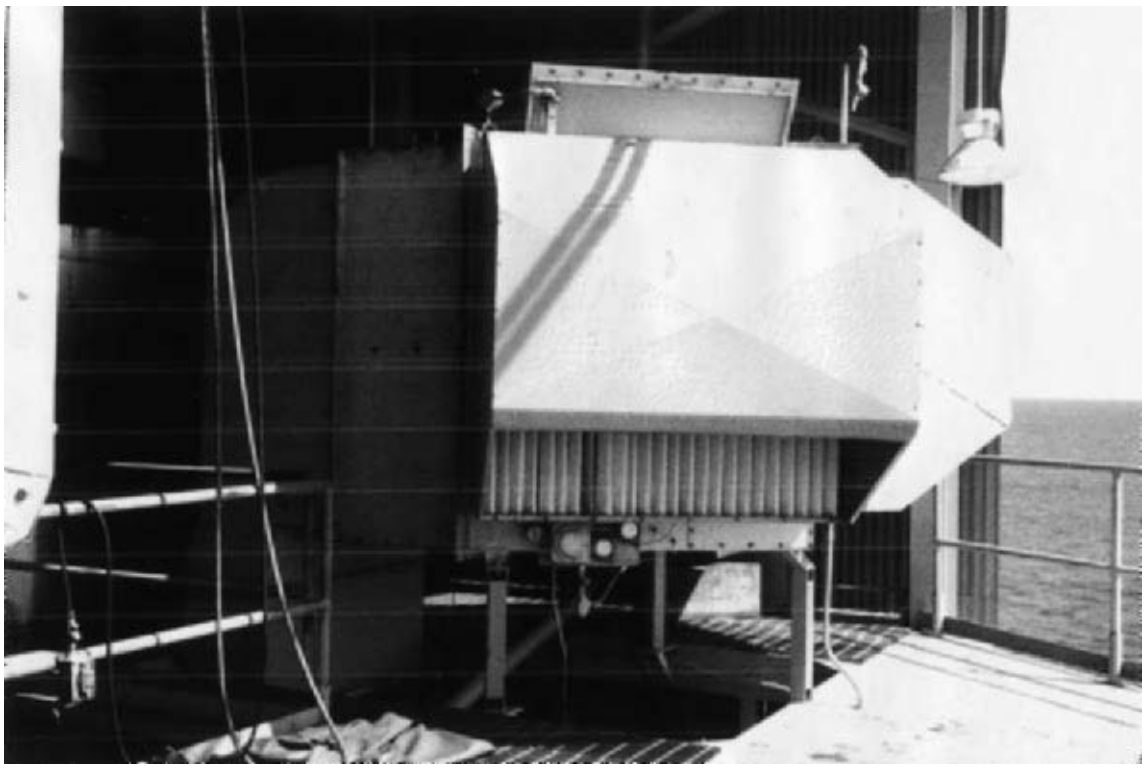


FIG. A-40 A filter housing with weather louvres and a bypass door. (Source: Altair Filters International Limited.)

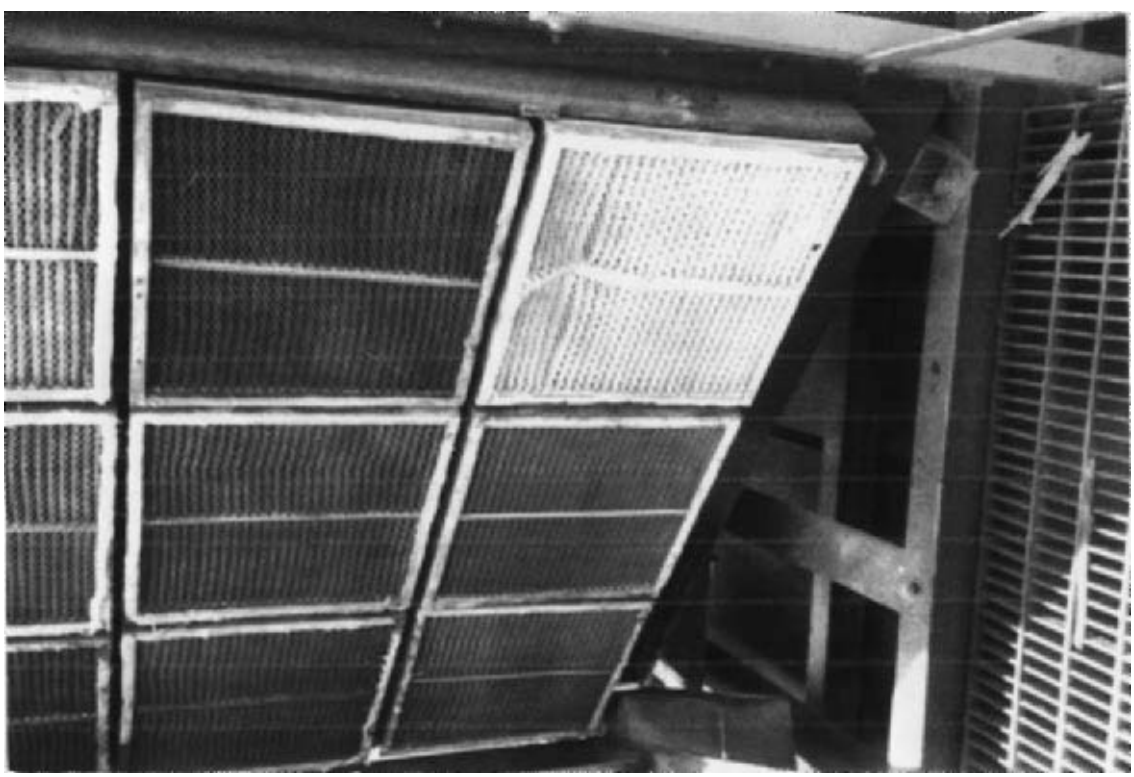


FIG. A-41 Typical high-efficiency cartridges. (Source: Altair Filters International Limited.)



FIG. A-42 A fouled compressor. (Source: Altair Filters International Limited.)

Fig. A-42 A fouled compressor rotor from an engine in Brunei.

Fig. A-43 Turbine corrosion from an engine in India.

Fig. A-44 Compressor fouling and corrosion from an engine in Indonesia.

Fig. A-45 Turbine blade failure from an engine in Indonesia.

Fig. A-46 Debris in an engine compressor in Brunei.

In addition to these visual indications, there are many instances where engine overhaul cost has soared because the installed filter system was ineffective. The main problems can be categorized as follows:

- Design suitability
- Material suitability
- Maintenance

Design suitability

The main problem with most of the systems was that they were incapable of dealing with the rainstorms that frequent the area. As previously shown, these storms are much more severe than in the more temperate parts of the world where the systems are normally designed.

In general, it was found that even where weatherhoods were fitted, they did not provide adequate protection. In Fig. A-40 the air immediately contracts and turns through 90° within a very short distance of the filter section. In addition, the weatherhoods induce an upward inlet airflow, with the result that the majority of

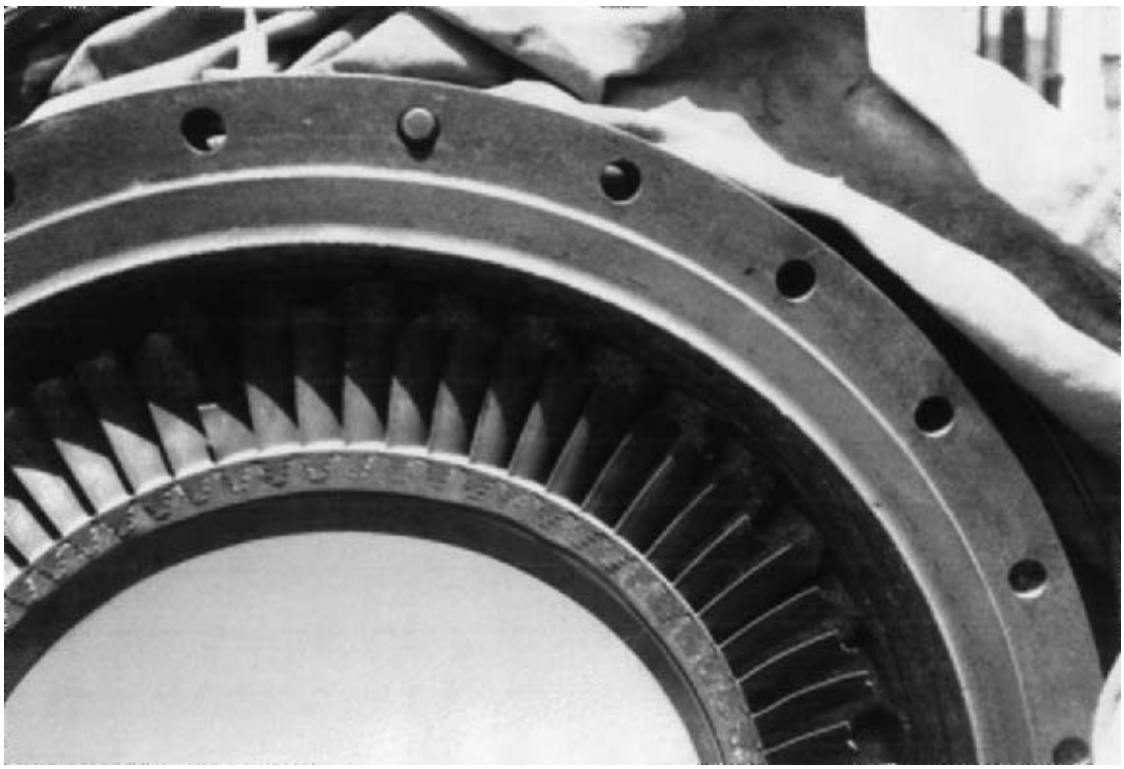


FIG. A-43 Turbine corrosion. (Source: Altair Filters International Limited.)

the airflow is concentrated at the lower half of the filter face, increasing the local velocity. Any water that is caught by the weather louvres drains vertically downward right into the area where the high velocities exist, with the result that the water is reentrained off the vanes into the filters downstream. This situation is further worsened as no drains were provided anywhere in the housing, including the weather louvre section. So even if rain had been caught by the weather louvres, it would have inevitably passed downstream.

The weather louvres that were mostly fitted were a low-efficiency type, with a large louvre pitch. Most were of a plastic construction that tends to embrittle and fracture with time. Their effectiveness against rain is so low that many operators were forced to shield them in some way. Figure A-47 shows a typical modification while Fig. A-48 shows a more elaborate protection. The latter solution is a very unwelcome compromise since the enclosed canopy is a potential gas trap.

On some of the more recent installations of this kind an attempt was made to provide drainage. Synthetic rubber dump valves were used, which rely on the weight of a column of water opening up a slit in the bottom of the valve. Unfortunately, the atmospheric pollution and the high-humidity climate tends to glue the valve openings. It is only when they are manually opened that the valve will discharge the water (Fig. A-49); at other times water will be reentrained into the filters.

Although great attention has been paid to designing systems that can remove large quantities of dust, in practice only a small percentage of sites in this climate can be regarded as dusty. The inclusion of inertial-type systems is unnecessary and too expensive. In addition, it actually worsens the already poor rain protection.



FIG. A-44 Fouling and corrosion. (Source: Altair Filters International Limited.)

Spin tubes, that employ high tangential air velocities, will shatter any incoming rain droplets into a fine mist, which then penetrates the remainder of the filter system. Similarly, the other inertial dust systems will do the same, although to a lesser extent. Any water that passes into the high-efficiency stage of the filters is potentially very serious, since it can wash contaminants through the filter and into the engine. If the installation is coastal or offshore, there is a real risk that sea salt may pass straight into the engine. Under these conditions the filter is a potential "salt bomb," waiting to explode as soon as the weather changes.

It has now been recognized that, in offshore or coastal environments, special attention is needed to prevent salt ingress. It is not sufficient just to provide a high-efficiency filter system without understanding the behavior of salt and making allowance for it. Salt, more commonly sodium chloride, will absorb water from the atmosphere under certain conditions of relative humidity. A seawater droplet will dry out at relative humidities of 40% and lower, but will absorb water from the air at high humidity. Therefore, if salt is caught on a filter and the humidity rises, the salt may absorb water and droplets will form that can then reentrain into the airstream and into the engine with disastrous results.

It was a combination of poor rain protection and salt carryover that caused disastrous problems to two Avon engines operating in offshore Brunei. Water and

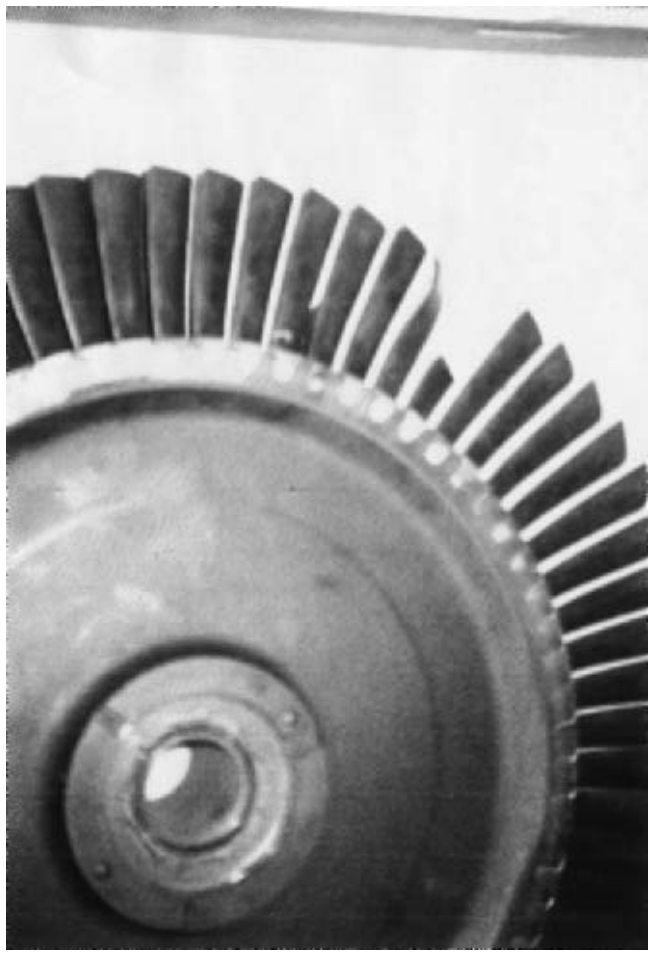


FIG. A-45 Turbine damage. (Source: Altair Filters International Limited.)

salt leached through the final-stage, high-efficiency cartridge filters (Figs. A-50 and A-51) and were carried into the downstream ducting and onward into the engine, causing massive compressor and turbine corrosion.

This situation was further exacerbated by the choice of materials for the inlet system. While the filter housing was constructed of stainless steel, the downstream silencer and plenum used a stainless-steel perforated lining for acoustic attenuation, welded to a carbon steel base frame.

The entrained water soaked through the perforated steel into the acoustic lining where it acted as a perfect electrolyte to assist the galvanic corrosion which can take place between even relatively similar metals such as carbon steel and stainless steel.

Severe inlet corrosion was experienced (Figs. A-52 to A-54) to such a degree that in places the perforated sheet became detached from the base frame (Fig. A-55). This released considerable amounts of debris into the engine, causing even further damage.

The whole inlet systems are now being replaced, which will involve a two-week shutdown on each of these turbines, with the associated loss in production. In addition, the overall costs of one turbine alone was in the region of US $600,000. An expensive exercise indeed.



FIG. A-46 Debris in a compressor. (Source: Altair Filters International Limited.)

Corrosion in the tropical environment is widespread. The initial use of carbon steel gives a limited life, particularly offshore. Figure A-56 shows the remains of an inertial filter still in service and faithfully painted by the operators. Figure A-57 shows the bottom of a corroded weather louvre section that indicates lack of damage. Figures A-58 and A-59 show severe corrosion of ducting downstream of the air filters, in which the provision of air filters becomes meaningless with such large air bypass routes. Figure A-60 shows the corroded duct downstream of the air filters, with surface debris sitting awaiting ingestion into the turbine. Clearly, corrosion is a massive problem.

The provision of filter bypass doors is a mixed blessing. These doors are notoriously difficult to seal, and are a potential area where all the money spent on a good filter system can be wasted. They are designed, of course, to help maintain production and are, at times, hazardous to the health of the turbine. A typical

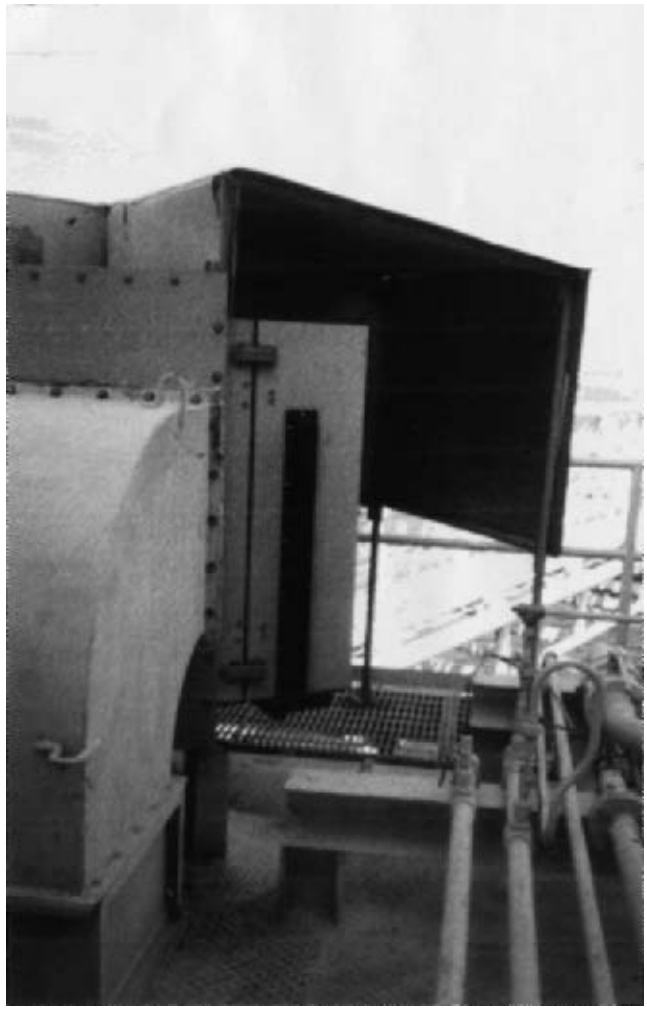


FIG. A-47 Modified weather protection. (Source: Altair Filters International Limited.)

example is when shot-blasting work is undertaken nearby. The filters quickly became blocked and the bypass doors opened, allowing the very erosive shot-blast material into the engine.

Modern engine-control systems now make bypass doors obsolete. A double-pressure-switch arrangement is used, one set to give an alarm and another set at a higher depression to initiate turbine shutdown. Occasionally, a third pressure switch is added (normally set for shutdown) but linked in so that two of the three pressure switches must activate before shutdown is initiated. This helps to prevent a spurious trip due to instrument error.

Painted carbon steel needs constant maintenance and, increasingly, sophisticated paint systems that ensure that the cost difference with stainless steel is reducing all the time. Figure A-61 shows a recently installed filter house where the initial paint has not been applied correctly, resulting in large bubbles on the metal surface. Figure A-62 shows part of a new carbon-steel filter housing awaiting installation, while Fig. A-63 shows that corrosion has already started.

It is not uncommon for filter housings to require repainting every two years to keep them in good condition. The painting process brings all the horrors of shot



FIG. A-48 Filters protected by a canopy. (Source: Altair Filters International Limited.)

blasting. The lifetime maintenance cost of a painted carbon-steel structure vastly outweighs the small extra premium that stainless steel affords for the initial installation.

Newer Filtration Designs

Clearly, the most worrisome deficiency of the old systems was the combination of poor water-handling ability either from rain or salt-related droplets and the materials of construction.

Gas turbines that operate on warships also experience intermittent heavy-water loadings. These are protected by vane separators, which are, in essence, a high-efficiency weather louvre. The main differences are the close vane pitching, which is 3 to 4 times closer than for weather louvres, the more intricate profile of the vanes, and the integral drain system. These vanes have a high capacity and efficiency and can easily cope with even the record rainfall rates mentioned previously without the need for a weatherhood, thereby saving weight and expense. Since droplets can develop on the filter stages due to humidity, a vane separator is also necessary as the final stage to prevent the problems shown in Fig. A-51.

In addition, technology has improved, so that new, tougher filter materials are now available that will allow filters to work at slightly higher velocities and so reduce air-filter housing sizes with the associated reductions in capital and shipping costs. These tough materials can withstand shot and grit blast without damage. The higher velocities allow the systems to be designed such that droplets will break off the filter stages but then be caught in the final-stage vane separator.

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